Substituting Reference σ-Profiles Degrades Solubility Prediction; on Activity Data, Bounding a Deployed COSMO-SAC Against Its Inputs Returns a Map, Not a Verdict

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Abstract

Physics-informed models route predictions through interpretable physical quantities, expecting accurate references to improve accuracy. We measure that for solid–liquid solubility: a graph neural network predicts charge-density profiles; fixed COSMO-SAC turns two into a solubility; a quantum-chemical database tabulates them. Grounding names two opposite-signed operations: supervising the profile against it during training lowers solubility mean absolute error by 0.20 (three seeds, not certified leak-free); substituting the reference for the learned profile at inference raises it, by 0.18 in the control removing the test-time swap (one seed) and 0.41 without it (three seeds). Neither resolves against the 0.20. On infinite-dilution activity data the same reference bounds the deployed thermodynamic model's error from below, its inputs' from above. The input bound barely moves between strata—an estimator property, not chemistry—while the error does, so the output is a map: 59 strata, 15 boundable, not one verdict. One admissibility rule leaves one row set standing—the glycol-ether solvents, 182 measurements, 43 pairs, where that model's own error exceeds its inputs' by at least $+2.04\,(\ln\gamma_2^\infty)^2$ (90% interval $[+1.27, +2.83]$)—and no other stratum, neither set whole, nothing on the solubility axis. It survived a 59-fold search: chemistry-blind relabelling certifies some row set in 58% of draws, reaching +2.04 in 10%. A pre-declared out-of-sample test returned *not testable*: the one independent database holds no glycol-ether solvent with an organic solute. Trained through this fixed closure, the learned profile departs from its reference along a shared direction (separate three-seed runs).

1 Introduction

How much of a solid dissolves in a given liquid decides whether a drug can be formulated as a tablet or an injection, which solvent a crystallisation is run in, and how a product is purified. It must be measured one solute, one solvent and one temperature at a time, so most industrially relevant combinations have never been determined, and independent determinations of the same system disagree by enough to floor what any predictor can achieve. Dissolving a solid also trades the cost of breaking up a crystal against the cost of mixing the freed molecules into the solvent, so a predictor has to be right about two unrelated kinds of chemistry at once, on molecules unlike any it was shown.

Predictors divide by how much physical structure they impose. Direct models regress solubility from descriptors or from a representation learned off the molecular graph; FastSolv[1] does so with a deep neural network and leads on extrapolation to new solutes. They are the more accurate there and they are opaque: the prediction arrives as a single number, with no account of which part of the chemistry it captured. Physics-informed models instead predict the thermodynamic quantities the physical picture names—an activity coefficient for the solute–solvent interaction, a melting temperature and heat of fusion for the crystal—and combine them through a fixed equation, as SolProp[2] does with a thermodynamic cycle of separately learned components. What that buys is decomposability: each part is a quantity chemists tabulate on its own, so it can be supervised against reference data a direct regressor has no way to use. On accuracy the black box leads and the physics-informed cycle trails it. Learned surrogates for the activity coefficient have advanced quickly, and split the same way. Winter et al.[3] predict $\ln \gamma_2^\infty$ from SMILES with a language model and Sanchez-Medina et al.[4] embed a Gibbs–Helmholtz constraint in a graph network for its temperature dependence, both making the activity coefficient the regression target; a second line instead keeps a mechanistic model and learns its input. COSMO-SAC[5] is a segment-activity reformulation of Klamt's COSMO-RS, and the σ-profile it consumes—screening-charge density over a molecule's cavity—is that framework's construct, tabulated from quantum chemistry.[6] Predicting it from structure removes that calculation from the pipeline; TeNNet-SAC[7] does so while also learning the segment-activity kernel, and gains from truer profiles.

When a fixed physical structure improves or degrades a learned model is the broader question of physics-informed machine learning,[8] whose dominant forms impose structure as a differentiable object: a governing equation enforced through the loss,[9] gray-box hybrids composing learned components with fixed mechanistic maps,[10] and operator learning with a trainable correction to a misspecified equation.[11] The prevailing expectation is that more physics improves generalisation, yet the same structure has documented failure modes in which it makes optimisation or accuracy worse:[12] in neural PDE solvers, a learnable term added to a fixed discretised operator absorbs truncation artifacts rather than physics, so the apparent interpretability is illusory[13]—though that error source is numerical and has no external reference. The same concern appears in concept-bottleneck models as *concept leakage*, a learned concept silently encoding information beyond its nominal meaning.[14–16] Recent work traces one route to leakage to the downstream model: a misspecified head forces the encoder to deform the concept into carrying a dependence the head cannot represent,[17] and overwriting a frozen head's leaked concepts with ground truth can then degrade accuracy.[18] The gap between a fixed physical map and reality is the model-discrepancy problem of Kennedy and O'Hagan[19] and its calibration critique;[20] four further literatures supply pieces used below—forecast decomposition, which splits a fixed predictor's error as §2.1.1 does; quasi-maximum likelihood, which gives the pseudo-true limit a misspecified closure converges to; identifiability, for when a composed model's parts are separable at all; and the validity of composed pipelines, for what a bound on one component licenses about the whole (§S12).

The design rests on an assumption rarely stated because it is taken for self-evident: a model supplied with physical inputs closer to their true values predicts better. For one class of intermediate that can be tested outright, because a reference value is tabulated and can be handed to the model in place of the one it computed. It needs testing, because when a predictor of this shape is inaccurate either the fixed physical model is wrong or the learned quantities feeding it are uninformative, and the two call for opposite remedies. What this literature lacks is a way to *measure*, rather than diagnose, which of the two limits a composed predictor's accuracy, and whether the learned intermediate has stopped reporting the quantity it is named after; in the concept-bottleneck case no external reference exists against which either question could be settled. This work supplies one—a reference σ-profile for solubility, tabulated substituent constants for a second chemical domain—and holds the mechanistic kernel fixed, so the tabulated value stays available as an external reference for the learned one.

Here a graph neural network predicts the pair of charge-density distributions COSMO-SAC consumes, in place of the quantum-chemical calculation that ordinarily supplies them. The tabulated counterpart can enter at either of two points: as a target the network is trained toward, or as a replacement for what the network computed at the moment of prediction (Fig. 1). The thermodynamic model is differentiable, so both routes are open on a single trained network, and the tabulated values are put to a third use, dividing the fixed model's error between the map and the inputs handed to it. Only one route helps. Whether the fixed map is what makes the other one fail is a separate question and is asked on separate data, because solubility data cannot say which component is at fault, one measurement constraining the pair jointly; activity coefficients can, because both molecules carry a tabulated distribution and the two error terms separate. Where they separate, an ordering decides the remedy: a map whose own error stands above what its inputs fail to carry does not become more accurate when handed a better input, but more exposed, the accurate input no longer masking the bias the map introduces. That ordering is a property of the chemistry the map is asked about, so the instrument returns a map over chemical strata rather than a verdict. Two consequences follow. A learned intermediate trained through a misspecified map cannot be read as the physical quantity it is named after: the predicted distribution drifts away from the tabulated one along a direction shared across molecules. And substituting a reference value for it at prediction time is not a safe improvement but a change of regime; here it degrades prediction at every seed. Neither consequence is peculiar to charge distributions: tabulated substituent constants behave the same way inside a fixed Hammett relation for acid dissociation, improving the aromatic series that relation was built for and degrading the ones outside it.

2 Computational methods

2.1 The composed predictor and its error split

For a solute s in solvent v at temperature T, the target is $y = \ln x_2$, the log saturated mole fraction. Dissolving a solid trades breaking the crystal lattice (Φ) against mixing the freed molecules into the solvent ($\ln \gamma_2$), and solid–liquid equilibrium (SLE) sets the log-solubility to their negative sum:

$$\ln x_2 = -\underbrace{\frac{\Delta H_{\text{fus}}}{R}\Big(\tfrac{1}{T} - \tfrac{1}{T_m}\Big)}_{\Phi(T)\ \text{(crystal)}} - \underbrace{\ln \gamma_2(x_2, T)}_{\text{activity}}, \quad (1)$$

with the constant-ΔC_p correction omitted for clarity. The activity term is produced by a *differentiable closure* g: a graph encoder predicts a σ-profile—the distribution of screening charge density over a molecule, the histogram a COSMO model consumes—for each molecule, and COSMO-SAC,[5] a segment-activity reformulation of COSMO-RS, maps the pair of profiles to $\ln \gamma_2$. The segment-interaction energy inside that map—the electrostatic-misfit and hydrogen-bonding terms setting what two surface segments cost each other—is the closure's *kernel*, the part the published revisions rewrite, so a closure variant is named for the year of its kernel: the deployed *2002 kernel*, and the *2010 kernel* that types segments as hydrogen-bond donors or acceptors instead of treating hydrogen bonding with one parameter. Write $z^\star$ for the *reference* physical latent, the σ-profile tabulated in the VT-2005 database,[6] and $\hat{z}$ for the head's own prediction. Feeding $z^\star$ in place of $\hat{z}$ (a σ-oracle) isolates the closure from encoder error: if the closure were well specified, $g(z^\star)$ should be the best available activity estimate. Each trained configuration, and each closure variant evaluated, is an *arm*; a *Direct-GNN* control arm drops the closure and emits $\ln x_2$ directly, and the two were tuned separately (§2.4).

Everything below is stated for a general *composed predictor* $\hat{y} = g(h(x))$ in which a fixed closure g consumes a learned physical interme-

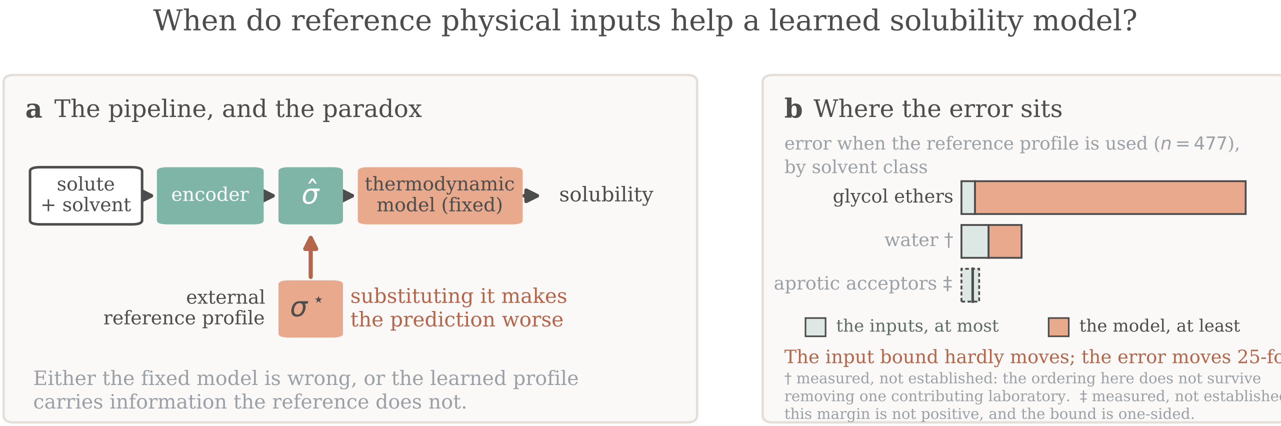

Figure 1: **Overview.** **(a)** A learned encoder predicts a charge-density profile $\hat{\sigma}$ that a fixed thermodynamic model turns into a solubility; the arrow marks the evaluation-time substitution of the reference $\sigma^{\star}$, which makes it worse. **(b)** The counterpart error on infinite-dilution activity data, by solvent class; only the unmarked class stands, each mark carrying its own reason, and each block is a one-sided bound rather than an interval. Training-time supervision is not drawn.

diate $z = h(x)$, against a matched free predictor that shares the encoder and drops g; solubility is one instance and the pKa closure of §3.6 a second.

The crystal/activity split of Eq. (1) is not identifiable from solubility alone. At infinite dilution with $\Delta C_p = 0$ the observable is affine in $1/T$, and the activity intercept and slope already span that plane, so the profiled Fisher information for ΔH_{fus} is exactly zero and the indeterminacy set is two-dimensional; at finite dilution the composition factor $(1 - x_2)^2$ leaves an information of order x_2^2. The deficiency closes only under a rank-completing external constraint—a direct crystal label, or a fixed activity slope (Lemma S1, §S2). That is why an under-determined factor needs external single-component data and why compensation between the factors is expected; it says nothing by itself about whether grounding helps.

2.1.1 The closure–insufficiency decomposition

Let m be the closure's target—here the experimental infinite-dilution activity coefficient (IDAC) $\ln \gamma_2^{\infty}$, in §3.6 the measured pK_a—and $g(z^{\star})$ the closure evaluated on the reference latent.

Lemma 1 (Exact closure–insufficiency decomposition and its one-sided bounds)**.** *For a fixed (non-fitted) closure g, with $z^{\star}$ denoting the closure's* entire *argument (the profile pair together with the temperature wherever the closure reads one),*

$$\underbrace{\mathbb{E}\big[(m - g(z^{\star}))^2\big]}_{\textit{ref.-input MSE}} = \underbrace{\mathbb{E}\big[\mathrm{Var}(m \mid z^{\star})\big]}_{B_{\text{insuff}}\ (\textit{irreducible})} + \underbrace{\mathbb{E}\big[(\mathbb{E}[m \mid z^{\star}] - g(z^{\star}))^2\big]}_{B_{\text{closure}}\ (\textit{decoder bias})}, \tag{2}$$

the cross term vanishing identically. Moreover $B_{\text{closure}} \geq (\mathbb{E}[m] - \mathbb{E}[g])^2$, and $B_{\text{insuff}} \leq \mathbb{E}[\mathrm{Var}(m \mid \mathrm{bin}(g(z^{\star})))]$ for any binning of $g(z^{\star})$.

Proof. (a) The identity, and the cross-term condition. $g(z^{\star})$ is $\sigma(z^{\star})$-measurable by construction, since $z^{\star}$ is the closure's whole argument. Split the residual as $m - g(z^{\star}) = [m - \mathbb{E}(m \mid z^{\star})] + [\mathbb{E}(m \mid z^{\star}) - g(z^{\star})]$, square and take expectations; the squared terms are B_{insuff} and B_{closure}, and the cross term vanishes by the tower rule, $\mathbb{E}[(m - \mathbb{E}(m \mid z^{\star}))(\mathbb{E}(m \mid z^{\star}) - g(z^{\star}))] = \mathbb{E}[(\mathbb{E}(m \mid z^{\star}) - g(z^{\star}))\,\mathbb{E}(m - \mathbb{E}(m \mid z^{\star}) \mid z^{\star})] = 0$, because the second factor is $\sigma(z^{\star})$-measurable. **That measurability is the condition, and it fails if any argument of g is left out of the conditioning variable**: with $g = g(z^{\star}, T)$ and T

excluded, the cross term need not vanish and the binning in (c) is no longer a coarsening of the conditioning σ-algebra, so neither the identity nor the upper bound survives. Where g reads a temperature—the broad IDAC set of §3.5.1, dim $=$ 103—we accordingly take $z^\star = (\text{profile pair}, T)$; on the VT-2005-matched set every row sits at 298 K and the two readings coincide. The same rule fixes $z^\star$ in the second domain: the Hammett closure of §3.6 reads a per-series constant pair as well as the substituent constant, so there $z^\star = (s, \sigma_{\mathrm{H}}^\star)$ and the conditioning is on both. *(b) Jensen lower bound.* With $\Delta(z^\star) = \mathbb{E}(m \mid z^\star) - g(z^\star)$, $B_{\text{closure}} = \mathbb{E}[\Delta^2] \geq (\mathbb{E}[\Delta])^2 = (\mathbb{E}[m] - \mathbb{E}[g])^2$, using $\mathbb{E}[\mathbb{E}(m \mid z^\star)] = \mathbb{E}[m]$. It needs *no* smoothness assumption; its one requirement is that m and g share the $\ln \gamma_2^\infty$ reference state, which the closure validation of §2.3 establishes. *(c) Law-of-total-variance upper bound.* For any binning $B = \text{bin}(g(z^\star))$ one has $\sigma(B) \subseteq \sigma(z^\star)$, and conditioning on a coarser σ-algebra cannot reduce expected conditional variance: $\mathbb{E}[\text{Var}(m \mid B)] \geq \mathbb{E}[\text{Var}(m \mid z^\star)] = B_{\text{insuff}}$. *(d) The estimand is convention-free; the bound is not.* $B_{\text{insuff}} = \mathbb{E}[\text{Var}(m \mid z^\star)]$ is a functional of the law of $(m, z^\star)$ alone and does not involve g, so it is identical across closure conventions and any $\text{MSE}(g_1) - \text{MSE}(g_2)$ is entirely $B_{\text{closure}}(g_1) - B_{\text{closure}}(g_2)$. The bound of (c) is not convention-independent, and no contradiction arises: two conventions coarsen the same $\sigma(z^\star)$ along two different statistics, giving two valid upper bounds on one quantity, of which the minimum is again valid but is biased low in finite samples, so we report it and do not headline it. $\square$

B_{closure} is the systematic discrepancy of a fixed physical map in the sense of Kennedy–O'Hagan,[19,20] and its dominance over B_{insuff} is the empirical content of the claim that the ceiling is set by the closure rather than by the inputs. **What B_{closure} is the error of is a composite object.** The identity compares $g(z^\star)$ against $\mathbb{E}[m \mid z^\star]$, and $z^\star$ is not measured but produced by a procedure—for σ-profiles a quantum-chemical calculation on one conformer and one protonation state per molecule at one level of theory—so B_{closure} is a joint property of the map and of the procedure that produced the reference it reads, and not of the kernel's functional form alone: a displaced reference loads onto B_{closure} through the curvature of g even where g is exactly specified (§S9). The deployed pipeline consumes that same composite, so it is the object a practitioner faces, but every attribution below is to the pair and not to the kernel. Where model-discrepancy calibration *fits* a correction from held-out observations, here g is fixed and $z^\star$ supplied from outside the fit, so the composite's error is *attributed* rather than corrected—available only where an external latent oracle exists. **Exactness is contingent on g being fixed**: for a *fitted* decoder the cross term does not vanish and a plug-in point split is invalid. Because B_{insuff} is bounded above and B_{closure} below, a positive *separation margin* $\text{MSE} - 2B_{\text{insuff}}^{\text{up}}$ establishes that the closure's own error exceeds what input insufficiency can account for, while a non-positive one is a failure to separate and licenses nothing: a reversal would need a *lower* bound on B_{insuff}, which this instrument does not supply. **Its size is a lower bound and not an estimate.** Since $\text{MSE} = B_{\text{insuff}} + B_{\text{closure}}$ exactly, $B_{\text{closure}} - B_{\text{insuff}} = \text{MSE} - 2B_{\text{insuff}}$, and the reported margin under-states that difference by $2(B_{\text{insuff}}^{\text{up}} - B_{\text{insuff}}) \geq 0$, an amount this design cannot measure. Every margin below therefore says the closure's error exceeds its inputs' by *at least* the number printed, and a bootstrap interval on a margin is an interval on that bound: its upper endpoint limits nothing.

What population, and what sets the resolution. The population is the *measurement superpopulation*: a draw is a nominal compound pair *at a temperature*, together with the conformer, tautomer and protonation state actually populated there, and an independent determination of $\ln \gamma_2^\infty$ for it. Neither the conformational state nor the replicate is part of the conditioning variable, so $\text{Var}(m \mid z^\star)$ is the spread of $\ln \gamma_2^\infty$ across the conformational ensemble and across replicate determinations at one reference profile and temperature—non-zero for physical reasons unrelated to sample size. Experimental label noise is a *component* of it, not a term beside it: an independent zero-mean error on m enters $\mathrm{Var}(m \mid z^\star)$ directly, so it sits inside every bound reported here and is never added on. What the design cannot do is *observe* it, neither set holding two rows at a common argument. The bounds are therefore conservative in one direction only—were the superpopulation variance negligible the finding would be the stronger $B_{\text{closure}} = \text{MSE}$—so nothing reported can be an artifact of over-estimating B_{insuff}, no entry anywhere is the share of the error the inputs contribute, and a low cluster-bootstrap frequency P is loss of resolution and never evidence for the inputs. Resolution is governed by $\dim(z^\star)/n$.

2.1.2 The decomposition and the grounding gap

The decomposition is the measurement the conclusions rest on: assumption-light, needing neither a representable encoder nor a lossless bottleneck. Alongside it we report a second, purely observable quantity, the *grounding gap* $\mathcal{G} := R_{\text{orc}} - R_{\text{e2e}}$, the oracle risk minus the best end-to-end risk; the paradox is $\mathcal{G} > 0$. Under a representable encoder (A1) and a lossless bottleneck (A2) it is sandwiched $0 \le \mathcal{G} \le B_{\text{closure}}$, so $\mathcal{G} > 0$ would certify $B_{\text{closure}} > 0$ (Theorem S1, §S2). A1 is empirically doubtful—the encoder is transfer-limited, a linear probe dropping from train $R^2 \approx 0.76$ to test $R^2 \approx 0.45$—and A2 is unverifiable and physically dubious; when A2 fails, $\mathcal{G} > 0$ conflates closure misspecification with input insufficiency, a well-specified closure with a lossy latent giving $B_{\text{closure}} = 0$ yet $\mathcal{G} > 0$. We therefore read $\mathcal{G}$ as corroborating evidence and base the model-dominance conclusion on the decomposition.

The sign rule, and scope. Whether the *trained* physics model outperforms the *trained* black box at sample size n follows a sign rule. Writing each trained risk as an asymptote plus estimation variance, $\mathbb{E}[\hat{R}_{\text{phys}}(n)] \approx R_{\text{e2e}} + V_{\text{phys}}(n)$ and $\mathbb{E}[\hat{R}_{\text{direct}}(n)] \approx R_{\text{direct},\infty} + V_{\text{direct}}(n)$ with $V(n) \downarrow 0$, the *physics penalty at budget* n is $\mathcal{T}(n) \approx \Delta_\infty + [V_{\text{phys}}(n) - V_{\text{direct}}(n)]$, with $\Delta_\infty := R_{\text{e2e}} - R_{\text{direct},\infty}$ its *asymptotic* part.

Proposition 1 (Sign of the physics penalty)**.** *(i) $\mathcal{T}(n) < 0$ (physics helps) iff $V_{\text{direct}}(n) - V_{\text{phys}}(n) > \Delta_\infty$: the variance a constrained prior saves must exceed its asymptotic bias. (ii) If $\Delta_\infty > 0$ and the variance gap vanishes ($V_{\text{direct}} - V_{\text{phys}} \to 0$), there is a critical budget $n^\star$ beyond which $\mathcal{T}(n) > 0$ for all $n > n^\star$.*

Both halves are immediate from $\mathcal{T}(n) = \Delta_\infty - D(n)$ with $D := V_{\text{direct}} - V_{\text{phys}}$ (proof, and the limiting refinement in which $\Delta_\infty = B_{\text{closure}} - \mathcal{G}$ when the control is asymptotically Bayes, in §S2). The consequence used below is that a prior is predicted to hurt precisely in the mature-data, low-noise, well-covered regime. Each physics-versus-control difference reported here is a penalty at the budget actually trained, not an asymptotic one; there is no "optimal grounding" theorem, and the decomposition does not improve out-of-distribution accuracy. Its variance term $D(n)$ is what carries the semiparametric phenomenon in which an estimated propensity outperforms the known-true one:[21,22] a fitted nuisance can be the more *efficient* estimator even when its bias is no smaller, so a true-valued input need not give the lower total risk at finite n.

2.2 Data, splits, and labels

Solubility measurements come from BigSolDB 2.0:[23] 127,930 rows over 15,665 solutes and 221 solvents between 243 and 438 K, partitioned by Bemis–Murcko scaffold into 111,724 training, 8,103 validation and 8,103 test rows so that every test solute is unseen at training; a melting-point-independent miscibility/structure filter sets row membership, and crystal T_m, ΔH_{fus}, Hansen parameters and infinite-dilution activity coefficients $\ln \gamma_2^\infty$ are merged on as per-solute labels where available. A split's rows do not all carry the same label: of the 8,103 test rows, 5608 carry a solubility measurement and the other 2495 carry a melting point and no solubility, so every accuracy number below is on 5608 rows and says so (§2.4). Crystal supervision is scarce—the test and validation splits contain essentially no

Table 1: Terminology, and the two symbols that would otherwise collide. Each term is defined here and used below without re-glossing.

Term	Meaning in this manuscript
closure g	the fixed, non-fitted physical map from the intermediate to the observable: here COSMO-SAC, in §3.6 a Hammett relation
kernel	the segment-interaction energy inside COSMO-SAC; a closure variant is named for its kernel's year (*2002*, *2010*)
arm	one trained configuration, or one closure variant evaluated; *DirectGNN* is the closure-free control arm and *TGNNSolv* the closure-carrying one, abbreviated *TGNN* in Table 5
$z^{\star}$, $\hat{z}$	the closure's *entire* argument at its reference (tabulated) value, and the network's own prediction of it; a σ*-oracle* feeds $z^{\star}$ where $\hat{z}$ would go. In §3.6 $z^{\star}$ is a pair and only one of its coordinates is learned
grounding	two opposite operations on one reference database: *supervising* $\hat{z}$ against it during training, and *substituting* $z^{\star}$ for $\hat{z}$ at evaluation
B_{closure}, B_{insuff}	the error of the closure *as deployed*—the fixed map read at the tabulated intermediate, so a property of the map and of the procedure that produced that intermediate *together*, and not of the kernel alone (§3.5.3)—and what its inputs cannot resolve, Eq. (2); B_{closure} is bounded below and B_{insuff} above, so the comparison is one-sided
margin	$\mathrm{MSE} - 2B^{\mathrm{up}}_{\mathrm{insuff}}$, the MSE of Eq. (2)—the closure read at $z^{\star}$ against its own target m, so in squared m units, squared $\ln\gamma_2^{\infty}$ on the activity axis and not the solubility axis; positive establishes $B_{\mathrm{closure}} > B_{\mathrm{insuff}}$ and its size is a lower bound on $B_{\mathrm{closure}} - B_{\mathrm{insuff}}$ and not an estimate of it; non-positive is failure to separate and licenses nothing
$\mathcal{G}$	the grounding gap $R_{\mathrm{orc}} - R_{\mathrm{e2e}}$, renamed from Γ so as not to collide with the segment activity coefficient Γ_S
stratum, map	a chemically defined row set of the broad IDAC set, and the 59 of them the estimator is run on
boundable	$n \geq 40$, so the fixed cell's eight bins hold five rows each
admissible	boundable *and* sign-stable under deletion of each contributing publication, in all four unit $\times$ convention cells (§2.5)
full, deployed	the combinatorial convention: *full* carries the Staverman–Guggenheim term, *deployed* is residual-only and is what this pipeline runs
UD	the University of Delaware σ-profile database
labelled test rows	the 5608 of the 8103 scaffold-split test rows carrying a solubility measurement; the other 2495 carry a melting point only and are scorable in no arm
broad IDAC set	477 IDAC $\cap$ UD rows, 185 pairs, all temperatures, UD profiles
VT-2005-matched set	60 pairs at 298 K carrying a VT-2005 profile on both sides
σ	alone, the screening-charge-density profile; $\sigma(\cdot)$ with an argument, the generated σ-algebra; σ_{H}, the Hammett substituent constant, $\sigma^{\star}_{\mathrm{H}}$ its tabulated value under the rule of §3.6 and $\hat{\sigma}_{\mathrm{H}}$ the network's; σ_{η}, experimental label noise

measured ΔH_{fus}, the quantitative basis of the identifiability argument (§2.1). The external single-component data used for grounding are an open crystal-property pool (~15,000 melting points), the VT-2005 σ-profile database, and ThermoML infinite-dilution activity coefficients.[24] Two hazards the pipeline handles explicitly—melting-point unit detection, and the instability of the seeded scaffold split across pipeline versions—are recorded in §S9.

The label-noise floor, and the two functionals it is read with. Independent determinations of the same system disagree, and that disagreement floors what any predictor can achieve. On the raw BigSolDB rows, group by (solute, solvent, temperature to 1 K) and keep the groups reporting at least two distinct sources; within a group the disagreement statistic is the mean absolute deviation of $\ln x_2$ from the group mean. Aggregating that per-group statistic over groups requires a choice of functional, and the two standard ones differ by a factor of five, so both are reported: the mean over groups is 0.15 $\ln x_2$ units for the 3948 groups backed by two sources and 0.31 for the 349 backed by three, while the median over the same groups is 0.03 and 0.09. Collapsing exact-duplicate values first—the same measurement re-tabulated across compilations, which would otherwise deflate the floor—leaves the mean at 0.16 over 4261 groups. The scaffold-split MAE reported here (1.70 to 1.85) is above the larger functional by a factor of five, so label noise does not account for the accuracy ceiling under either reading. This floor is a mean absolute deviation in $\ln x_2$; the aleatoric estimate published for this task, an inter-laboratory RMSE of 0.75 in $\log_{10} S$ on 34 duplicate solutions,[1] is a different functional on a different base, and the two are neither pooled nor converted into one another anywhere below.

Disclosure: the scaffold guarantee is not certified for these runs. The released stream builder excludes any pool molecule whose scaffold appears in validation or test and aborts on a missing split file—but that is a property of the builder, not a certified property of the runs reported here. One σ-stream build on disk was generated against the wrong split files. Counted inside the stream, 91 of its 1425 rows are held-out solutes by canonical SMILES; counted where such a row can act, 7 of the 147 solutes spanned by the 5608 labelled test rows have their own reference profile in that stream and 8 have a scaffold relative in it, on 297 and 351 of those rows. Solute coverage, not stream-row fraction, is the load-bearing figure for a scaffold split. Per-run stream provenance was not logged, so the artifacts cannot certify which build fed which arm. Each fact that bounds a leak bounds one contrast: the paradox contrast is one checkpoint evaluated two ways, and DirectGNN uses no stream. The grounded-versus-ungrounded supervision contrast is bounded by neither, because the ungrounded arm carries no σ stream either, so the only arm a leak can reach there is the grounded one, and the 0.20 gain of §3.1 stays uncertified until the leak-free re-run. §S9 gives the file evidence, each fact with the contrast it bounds, and the one molecule the leak touches in the n=44 analysis of §3.3. We therefore do not assert the scaffold guarantee for these runs.

What the leak-free re-run reports, and what it decides. The re-run is specified here rather than described afterwards, so that its protocol is fixed before its numbers exist. It trains the ungrounded and the grounded arm at five seeds (42 through 46) under the COSMO-SAC configuration of the published arms, unchanged. The σ stream is rebuilt with the scaffold guard pointed at the validation and test files of *this* split, and is certified inside each training run rather than at the builder: the run re-reads the stream it is about to consume and the split files it will be scored against, and asserts that they share no molecule by canonical SMILES and none by Bemis–Murcko scaffold. The stream's SHA-256 and row count are written into that run's manifest beside the SHA-256 of each pinned split file, so the artifact states which build fed which arm; the ungrounded arm's manifest records that it carried no stream at all, since here the absence is the thing asserted. Accuracy is read on the same 5608 labelled test rows as every other number in this paper, under the same cross-arm intersection rule (§2.4), and the five per-seed MAE values are reported for each arm rather than a mean and a standard deviation alone. Two further readings are taken on those same checkpoints: the evaluation-only substitution of §3.1, which is within-checkpoint and therefore a consistency check rather than a test, and the n=44 profile comparison of §3.3 with the one molecule the leak touches removed. Quoting the magnitude of the co-adaptation control of §3.1 as something other than one seed would need a separate pass at the same five seeds, which this protocol does not commit to running.

The decision the re-run settles is stated in advance and in one direction only, since it is the supervision contrast alone that hangs on it. If the five per-seed gains are of one sign, the two senses of grounding keep their opposite signs, and the abstract, §3.1 and Table 4 print the re-run's per-seed values in place of the three uncertified ones, with the refusal to order the two operations by magnitude unchanged. If the per-seed gains straddle zero, supervision has no established sign at five seeds: the paper then reports one operation with a sign—

substituting the reference at inference hurts—and records the other as unresolved, and the abstract, the heading of §3.1, the opening sentence of the Conclusions (§4) and the supervision row of Table 4 are rewritten to say so. If the gains are of one sign and it is the opposite one, that is what is reported. In either of the last two cases the graphic for the table of contents, which prints the two senses of grounding and their opposite signs, is redrawn to what is then reported, and the freeze contrast of §3.3 is restated as well, because the warm-up state whose quality it turns on is the supervised one. The substitution result is unaffected in all three cases: it is one checkpoint evaluated two ways, and no σ-stream build can reach it.

Which displays the five-seed values replace is fixed here as well, and in all three branches alike, because a five-seed abstract printed beside a three-seed figure would be a contradiction the manuscript had left itself the choice of. Three arms carry those values: the ungrounded arm, the grounded arm, and the σ-oracle arm, which is the grounded checkpoint re-evaluated with $z^\star$ substituted and therefore moves with it. Each display named next carries at least one of the three and is regenerated from the five-seed arms by the script that produced it—the abstract's 0.20 and 0.41; §3.1 entire, together with the two sections the restructure of 2026-08-07 moved its prose into—§3.2, which carries the regression slopes read off Fig. 3, the ranking contrast and the solute-blind comparison, and §3.4, which carries its channel paragraph—the per-seed gains and the seed-42 values included wherever they now print; the five rows of Table 4 under *the two senses of grounding* and the +0.14 of its accuracy row; the seed list of the σ-grounding row of Table 3; the ungrounded, grounded and oracle bars of Fig. 2 with their R^2; panels (b), (c) and (d) of Fig. 3; the block-1 grounded and σ-oracle rows of Table 5 together with the clip sweep in its caption, the fuller sweep of §S4 and the readings §3.7 takes from both; Table S2 and the unrounded means, per-seed values and arm orderings §S3 reads off it; Table S17 with the ranking and per-group recalibration readings §S6.1 sets around it, whose two columns are the grounded checkpoint and its own σ-oracle re-evaluation and nothing else; §S6.4 with Tables S19 and S20, which resolve that same +0.14 along chemical axes and restate the two arm means it is taken from; the row-class split of the substitution in §S1, whose −0.000, +0.431 and +0.006 put the +0.41 on the solvent channel; and the graphic for the table of contents. Four of those floats—Figs. 2 and 3, Tables 5 and S2—then carry re-run arms beside arms that were not re-run, so each states in its caption which of its arms came from the re-run—Table 5's block header already carries the form of words for it, and Fig. 3 remains one seed for all four of its panels, the new seed 42 in three of them and the published one in the control. Table S17 needs no such statement, both its arms being re-run; Tables S19 and S20 print no arm at all, every cell being a difference against the control, so each says in its caption that only the physics side is five-seed. Three numbers instead keep their published value with the five-seed one printed beside them and the caption saying so, because the re-run supplies one side of each comparison and not the other: the Staverman–Guggenheim contrast 1.8457 → 1.8788, whose combinatorial arm is not retrained; the channel swap's +0.27, a seed-42 checkpoint that is not retrained either; and the co-adaptation control's +0.18 together with the 0.59 share of §3.1, which move only if the separate five-seed pass above is run. Three passages carry no arm value but state that the 0.20 is uncertified—the disclosure above, limitation (xii) of §3.7 and the provenance paragraph of §S9—and the in-run certification answers all three, so each is rewritten to what the re-run certifies and limitation (xii)'s 0.20 moves with the abstract's. Nothing else moves, and the boundary is worth stating because two neighbouring numbers sit just outside it: neither the DirectGNN control nor the NRTL arm is retrained, so their rows in Tables 5 and S2, and the endpoint comparison of §S6.5—which is the NRTL arm against that control—stand as printed; and §3.5 uses no trained model at all.

2.3 Model, closure, and training

A *shared* message-passing encoder embeds solute and solvent, a bipartite cross-attention block lets them interact, and the resulting pair representation feeds every head; two interchangeable encoder alternatives do not outperform it. A fusion head predicts T_m and ΔH_{fus}, giving the ideal term $\Phi(T)$ of Eq. (1). The activity term $\ln \gamma_2$ comes from one of three differentiable closures. Two are controls, the fitted NRTL model and its symmetric one-parameter collapse. The third, parameter-free, is the closure under study: a σ-profile head emitting a 51-bin screening-charge-density profile per molecule, and a COSMO-SAC layer mapping the profile pair to $\ln \gamma_2$ with an optional Staverman–Guggenheim combinatorial term—the *full* versus *deployed residual-only* distinction on which §2.5 and §3.5 turn. The saturated mole fraction solves $\ln x_2 = -\Phi(T) - \ln \gamma_2(x_2, T)$ as a damped fixed point in x_2; a bounded, gated adaptive correction perturbs the physical parameters rather than the output, so the physical decomposition survives it.

External single-component data enter through *grounding streams*—the training-time sense of grounding. Each stream is a sidecar of *self-solvent* rows (solvent = solute, no solubility label, only the target factor's mask active), interleaved into training under the guard above; the streams ground the crystal factor from the melting-point pool and the activity factor from the σ-profile database, identifying from external data a factor that solubility alone leaves non-identifiable and that a black-box predictor cannot use. Training runs a three-stage curriculum under a weighted loss: a warm-up on the auxiliary streams alone, then joint training against solubility, then a low-rate stabilisation stage. The schedule pinned for the arms of Fig. 2 freezes the σ head at the end of the warm-up, so the σ stream enters the warm-up only.

The in-house closures, and where their validation stops. The dispersion-off closures in the fidelity measurement (§3.5.3) are our own differentiable implementations, so those arms share one code path, and each consumes its own model-prescribed σ-averaging (Mullins for 2002, Hsieh for 2010). Our 2002 layer—the *fair 2002* arm wherever the kernels are compared, because it reads all three profile channels—reproduces the NIST reference COSMO-SAC-2002[25] to RMSE 0.0035 $\ln \gamma$ on identical VT-2005 profiles, and our COSMO-SAC-2010/dsp layer,[26,27] built on the *typed* latent of three donor/acceptor-resolved σ-profiles per molecule in place of one, reproduces the NIST 2010 reference with dispersion *off*. The dispersion term is *not* validated to that standard, which is why every dispersion-off contrast is computed on our layers while the *2010, dispersion on* runs are quoted from `cCOSMO`, that reference implementation itself.

The same closure against a published evaluation. Reproducing a reference implementation fixes the code; it does not say whether the error this paper decomposes is the error a published COSMO-SAC has, or something peculiar to this data assembly. Table 2 answers that. De Souza et al.[28] score the original Lin–Sandler COSMO-SAC on the infinite-dilution activity-coefficient database distributed with *The Properties of Gases and Liquids*, at AAD $= \mathbb{E}|\ln \gamma^{\infty}_{\mathrm{calc}} - \ln \gamma^{\infty}_{\mathrm{exp}}|$ of 1.75 with HF-TZVP profiles and 0.68 with BP-TZVPD-FINE: one closure, two profile databases, a factor of 2.6 between them. An evaluation of COSMO-SAC is in that sense an evaluation of the profiles it was handed, which is why B_{closure} is defined above on the composite of map and profile calculation and not on the kernel. We ran the deployed `CosmoSacLayer`—its own class, constants and damped fixed point—over the same three record files, resolving components through the repository's CAS-to-InChIKey map and matching by exact InChIKey to this paper's VT-2005 artifact: 7889 of 8138 usable records (96.9%), 2279 distinct pairs, each at its own temperature. It lands inside the bracket the published closure spans across profile databases, on both statistics and under both conventions, and so do this paper's own two sets. Two limits stay open: the profile database is the one axis we

Table 2: The deployed closure against a published evaluation of the same closure. AAD is the mean absolute error in $\ln\gamma^{\infty}$.

COSMO-SAC-2002 on	profiles	n	AAD	R^2
As published[28]				
PGL 6th ed. IDAC set	HF-TZVP	6977	1.75	0.88
PGL 6th ed. IDAC set	BP-TZVPD-FINE	6977	0.68	0.96
The same set, through this paper's closure				
full convention[a]	VT-2005	7889	0.98	0.88
residual (deployed)	VT-2005	7889	0.77	0.92
non-aqueous rows	VT-2005	4347	0.47	0.68
aqueous rows	VT-2005	3542	1.13	0.83
This paper's own sets, residual convention				
broad IDAC set	UD	477	1.12	0.61
VT-2005-matched set	VT-2005	60	0.81	0.40

[a] The like-for-like row, a published COSMO-SAC carrying its Staverman–Guggenheim term; the residual row is the convention this paper's B_{closure} is measured in. Worst cell: water in an organic solvent, AAD 2.02 with a +1.92 bias on 468 rows. At 298 K and non-aqueous—the regime the n=60 set occupies—AAD is 0.42 on 1033 rows, against that set's 0.81, so this paper's set is if anything the harder one.

cannot match, VT-2005 being what this paper deploys and the axis that publication shows dominates; and the publication's 6977 points are not proven to be the same rows as the 7889 scored here, the record selection not being stated file by file. Subject to those, the closure error decomposed in §3.5 is the error a published COSMO-SAC has; which half of this paper's chemical localisation carries across to that set, on 130 times the data and someone else's compilation, and which half does not, is in §3.5.3.

2.4 Evaluation protocol, and the DirectGNN control

The *oracle substitution* replaces the predicted σ-profile $\hat{z}$ with the reference VT-2005 profile $z^{\star}$ before the closure, matched by canonical SMILES. Precisely: the σ-head emits a normalised 51-bin shape and a cavity area whose product is the area-weighted profile it passes to the closure, and for a matched molecule that pair—profile and area together—is replaced by the tabulated pair, which stands in the same relation, the reference area being the integral of the reference profile. So the reference area does reach the residual term's A_2/a_{eff} prefactor, and it reaches it as part of replacing one object rather than as a separate choice. Solute and solvent are matched independently. The size-factor volume is not substituted—the combinatorial term is switched off in these arms—and neither is the crystal head's output, which reads a solute-only embedding the substitution does not reach; the bounded correction does read the closure's own parameters, so the crystal values the solver finally receives shift by a little, at most 7×10^{-4} in Φ against a median magnitude of 4.7. §S1 lists every replaced and unreplaced quantity, and gives what the match rule reaches: the reference table covers 99.3% of the labelled test rows on the solvent side and 5.3% on the solute side, so the substitution contrast below is carried by the solvent profile.

It is applied three ways to rule out implementation artifacts: at evaluation only; injected at training time so the model co-adapts; and as a channel-swap control, injected at training under a coordinate-descent schedule that freezes the crystal branch during the SLE phase, so that only the activity branch refits against it. The three differ in when the replacement is made, not in what is replaced. The last two are at seed 42 only. An analogous override substitutes reference crystal targets. The activity target for the decomposition is the infinite-dilution $\ln\gamma_2^{\infty} = \lim_{x_2\to0}\ln\gamma_2$, on two sets that differ in their reference database and their regime (§3.5.1–§3.5.2).

DirectGNN shares the encoder, interaction, readout and pair representation, adds a thermometer (cumulative-threshold) temperature encoding, and maps directly to $\ln x_2$ with no heads, closure or solver. The two arms therefore share their representation but not their inputs or their tuning: the control receives an explicit temperature encoding the physics arm does not, seeing T instead through $\Phi(T)$ and the closure, and the optimisation settings were tuned per arm and differ by up to an order of magnitude. The TGNNSolv−DirectGNN gap is accordingly a comparison of two separately tuned pipelines that differ in the thermodynamic bottleneck, not an isolation of the bottleneck, and every physics-penalty comparison below carries that caveat (§3.7).

The controlled comparisons of §3.1 are run at three seeds and reported as mean±sd on a fixed row set; 5608 is that lock—the test rows that carry a solubility measurement, 5608 of the 8103 (§2.2)—so all arms score identical rows. The 2495 rows outside it carry a melting point and no solubility and are unscorable in every arm alike, the control included; none is dropped for anything an arm predicted. The code that builds the lock also asks for a finite $\ln x_2$ prediction in every arm, and that clause is inert: all twenty-two per-row prediction files are full length, and across them it removes no row, from any arm, at any seed. The set is fixed by which rows carry a label and not by which rows an arm handled; §S3.1 gives the decomposition. Table 3 names which run family produced which numbers, how many seeds it carries, and the script and artifact each regenerates from.

2.5 Estimating the split: one cell, one convention, one rule

Four analyst choices set the law-of-total-variance bound—the bin count, where the equal-count bin edges fall, the within-bin variance convention, and the combinatorial convention—and all four are fixed here, before any margin is reported.

The estimator cell. Eight equal-count bins of $g(z^\star)$, the *unbiased* within-bin variance, and the row unit, applied identically to every set and every stratum whatever its size. The bin count is the choice with the most leverage—it takes the broad set's aggregate margin from $+0.01$ to $+0.96$—so the cell is not left standing alone anywhere a finding rests on it: the same sweep is run for the strata the admissibility rule passes, and reported with them. Enumerating every equal-count contiguous partition, the bin-edge choice is immaterial on the broad IDAC set and material on the VT-2005-matched set, whose margin changes sign inside the envelope; we accordingly read that set's margin as consistent with zero and carry the claim on the broad set in stratified form.

The convention rule. The margin depends on the combinatorial convention, so one rule governs both sets: headline the *deployed* residual-only convention, pairing MSE and the bound computed under that same convention. Headlining the deployed convention on the smaller set and the full convention on the broad one would be the favourable choice on each set taken separately; one rule applied to both costs the broad set's margin about half its value. The Jensen constant-offset bound is convention-specific—it separates under the full convention and inverts under the deployed default—so what is reported rests on the model-free law-of-total-variance bound, which fits nothing and so lets no held-out row borrow strength from a near-twin. Both bounds are valid on the one population quantity, by Lemma 1(d), and so is the smaller.

The stratification rule. The broad set is stratifiable where the VT-2005-matched set's 60 pairs are not, on two axes that are pure functions of molecular structure: an ordered nine-class cascade of substructure tests over the solvent, hydrogen-bonding role first and dominant functional group second, and the solute's role as water or as an organic. The classifier reads a SMILES string and nothing else—never m, g, the squared error, the source or the bootstrap—so no cell can have been chosen on its outcome, which is a property of the code rather than an assurance. A stratum is *boundable* at $n \geq 40$, where eight bins hold five rows; the smaller ones are reported at that same cell, with the adaptive bin count $\max(3, \lfloor n/10 \rfloor)$ in the rows n it is applied to carried beside them in the deposited table, nothing being dropped for being small.

The admissibility rule. A stratum is *admissible* only if it is *(a)* boundable at that fixed cell and *(b)* robust to leave-one-source-out—its margin keeps its sign when each source publication contributing to it is deleted in turn—in all four unit $\times$ convention cells, since the map reports all four. That last clause is load-bearing and is never dropped: without it the set taken whole passes too. A stratum drawing on one publication fails (b) by non-testability, there being no deletion to perform; a deletion leaving fewer than 16 rows leaves the fixed cell undefined, and fails (b) as well. A third clause reduces the admissible *cells* to row sets, and it is stated here with the same precision rather than left to prose, because the one claim this paper carries from the map and the multiplicity null

Table 3: Run families: which runs produced which numbers, and at how many seeds. Scripts are in `scripts/analysis/`, artifacts under `results/`.

Run family (script)	Seeds	What it produced
σ-grounding arms, GPU, full corpus (`run_e5_comparison`)	$42, 43, 44$, mean±sd on the n=5608 labelled test rows	Figs. 2 and 3, Table S2, the +0.41 and −0.20 of §3.1
Co-adaptation and channel-swap oracle controls; gated output residual	42 only: a sign, not an effect size	the +0.18 and +0.27 of §3.1; §3.3
Decomposition, VT-2005-matched (`run_b_insuff_decomposition`)	none: the closure has no learnable parameters	§3.5.2; the estimator grids
Decomposition, broad IDAC (`run_b_insuff_representative`, `…_audit`)	none	§3.5.1: the +0.51, the +0.95, the deletion curve
The stratified map (`run_b_insuff_stratified_map`)	none	the 59 strata, Tables S10–S14
Closure-fidelity kernels (`run_fidelity_lever_fair2002`, `run_closure_flip_ci`)	none	§3.5.3, Table S7
Kernel residual (`run_local_closure_fix`)	five fit seeds; ± is their standard error	the $46 \pm 5\%$ and $-5 \pm 16\%$ of §3.5.3
Latent against its reference (`run_compensation_surrogate`)	$0, 1, 2$; mean±sd	§3.3, Fig. S4
Closure validation (`run_closure_reference_validation`, `run_published_idac_closure_check`)	none	the NIST reproduction; Table 2
External baselines (`run_external_baseline_comparison`)	one run each, on a *by-solute* split and, for two of the rows, a *random-pair* one	Table 5
pKa second domain (`run_pka_real_decomposition`, `…_hammett_probe`, `…_binsuff_boot`)	K=20 within-scaffold splits; G=4 series as the cluster unit	§3.6
Noise floor and encoder probe (`run_noise_floor_estimate`)	none	§2.2, §2.1.2

The seed-44 per-row oracle prediction file had been deposited at 295 of its 8103 rows, the transfer off the compute volume having stopped mid-record; the recovered complete file reproduces every metric in the run's own summary exactly and matches those 295 rows byte for byte (§S9), so every row of every arm at every seed is re-analysable from the deposit and the oracle row of Table 5 is a three-seed mean like the rest of its block. Fig. 3 is one seed, 42, for all four of its panels alike, and its caption says so.

that tests that claim both turn on it. *(c)* Admissible cells holding identical rows count once; and an admissible positive row set X is *demoted* to another, Y, when Y is admissible and positive, Y holds more rows than X, more than half of X's rows are Y's, and the residual $X \setminus Y$ is not itself admissible—so that X's margin does not stand outside Y. What decides it is that residual and not the overlap: a row set most of whose rows lie inside a larger one survives (c) whenever its own margin stands outside that larger one, so *(c)* is not containment and no cell is demoted for overlapping (§S3.2). Clause (c) takes the six admissible cells of §3.5.1 to three distinct row sets; positivity then removes one of the three and (c)'s residual test the second, leaving one. It is rebuilt on every draw of the multiplicity null, on the permuted labels, like the rest of the rule (§S3). Everything else is *reported in the map and not stated*, with its n, its sources, its margin, its interval and the reason. The rule governs the cells that favour us and those that do not alike, and it tests whether a number is stable, not what it means: an admissible stratum whose margin is not positive, and one that (c) demotes, both establish nothing.

3 Results and discussion

Cluster-bootstrap frequencies P are quoted as bands—rounded to one decimal, with > 0.95 and < 0.2 closing the ends—because with 17 solvent and 41 solute clusters on the VT-2005-matched set they do not carry two-decimal resolution; the two-decimal values are in the deposited artifacts. The resampling intervals of this paper—the Abstract's, the margins of §3.5 and the pKa intervals of §3.6—are *percentile* intervals: their endpoints are order statistics of a resampling distribution, with no bias correction and no studentisation, over the unit named beside each. §3.5 resamples solute and solvent clusters two-way, 3000 draws; in §3.6 the decomposition bounds resample molecules within each Hammett reaction series and the flip gap resamples the four series themselves, 4000 draws. Not every ± or bracketed span below is one of those. Two in particular are not: the kernel-transfer figures of §3.5.3 are a standard *error* over five fit seeds, and the leave-one-publication-out span of §3.5.1 is the range of that curve. Table 4 collects the load-bearing claims with the set each was measured on, its size, and the value with its uncertainty and scope.

3.1 Supervising the intermediate toward the reference, and substituting it at prediction time

The word "grounding" names two opposite operations on the same VT-2005 database. Supervising the σ head against the database *during training* improves solubility MAE from 2.043 ± 0.040 (ungrounded) to 1.846 ± 0.053 (grounded), a gain of 0.20 over three seeds. The ungrounded arm carries no σ stream, so this is the one contrast in the paper that the uncertified stream build of §2.2 could inflate rather than one the arguments there bound (§S9). Substituting the reference profile for the learned one *at inference* degrades it, and that substitution alone is the effect this paper is named for. It erases any positive R^2 (Fig. 2): the model predicts solubility better *from its own learned σ-profile* than when *given the external reference one* (MAE 1.85 vs 2.25, in $\ln x_2$ units). Of the figure's eight arms two are non-COSMO controls, DirectGNN and a fitted NRTL closure; five COSMO-SAC arms differ only in how the σ head is trained and in what it is fed, and the sixth (*grounded+comb.*) differs instead in the closure, restoring the Staverman–Guggenheim combinatorial term. *Ref. σ (eval)* is the grounded checkpoint re-evaluated with the reference profile substituted at test time, *ref. σ (train)* the co-adaptation control injecting it during training instead.

The headline +0.41 is the evaluation-only substitution, and part of it may be unrelated to the closure: swapping an input distribution at test time degrades a pipeline on its own. The control that removes that component injects the reference profile *at training*, so the model co-adapts; at seed 42 it costs +0.18 ($1.803 \rightarrow 1.981$) where the evaluation-only substitution costs +0.43 at that same seed ($1.803 \rightarrow 2.232$), so part of the headline gap goes with co-adaptation and the co-adapted arm is by design free of *that* confound. A channel-swap control costs +0.27, to 2.08. It carries a second confound the co-adapted arm carries too: neither is certified to have trained on the same σ-stream build as the seed-42 comparator both are read against, and the file times settle nothing either way (§S9). Both are worse than the learned-σ arm, which rules out an implementation artifact. Only the three-seed arms can be set against one another, and there the evaluation-only substitution costs about twice what supervision gains (+0.41 against −0.20). That ratio is between two three-seed arms and carries the co-adaptation confound with it, so it is not the paradox's co-adaptation-free magnitude; the arm that is, at +0.18, was run at one seed on an uncertified stream build and cannot be ordered against the −0.20 at all, whose own per-seed values (0.237, 0.075, 0.281) straddle it. Which of the two operations is larger is therefore unresolved, and only their signs are carried forward.

3.2 What the substitution costs: level, spread, and the choice of solvent

Figure 3 sets the same rows out predicted against measured, for the control and three of the closure arms. Two things are visible there that the arm ordering is not. First, every arm is severely shrunk toward the corpus mean: the ordinary least-squares slope of predicted on measured is 0.515 for DirectGNN, 0.416 ungrounded, 0.528 grounded and 0.690 with the reference profile, none near 1, and in every panel the median trace crosses the identity—sparingly soluble compounds are called more soluble than they are and highly soluble ones less. That shrinkage is not the closure's doing: the closure-free control's slope differs from

Table 4: One row per load-bearing claim: what is claimed, the data set and its size, and the value measured with its uncertainty and its scope. Notes *a*–*h* qualify a row or a column.

Claim	Set (n)[a]	Value, with its uncertainty and scope[b]
The two senses of grounding (§3.1)		
Reference profiles hurt solubility at evaluation (MAE)	labelled test rows (5608 of 8103), 3 seeds	$1.85 \to 2.25$; the per-seed values do not overlap; the gap carries the test-time input swap with it
…and still hurt when injected at training	labelled test rows[c], seed 42	$1.80 \to 1.98$ (own baseline $1.80 \to 2.23$); one seed and a stream build not certified against its comparator's (§S9), so a sign and not an effect size—free of the test-time swap, which the row above is not
…so part of that gap is test-time distribution shift	labelled test rows[c], seed 42	share 0.59 on one seed: a description of that seed, not a magnitude
Training-time σ supervision *helps* (MAE)	labelled test rows, 3 seeds	$2.04 \to 1.85$; the per-seed values do not overlap; per-seed gains 0.237/0.075/0.281 straddle the co-adapted arm's $+0.18$, so the two operations are not ordered by magnitude—and the only contrast here a σ-stream leak could inflate, the ungrounded arm carrying no stream (§2.2)
The evaluation substitution degrades the *choice* of solvent, not only the level	labelled test rows grouped by solute and T (823 groups, 107 solutes), 3 seeds	per-solute Spearman -0.197 $[-0.282, -0.117]$ pooled, all three per-seed intervals excluding zero; nDCG@3 -0.084 $[-0.124, -0.043]$; top-1 -0.107 $[-0.192, -0.020]$ pooled only, two of three per-seed intervals spanning zero. Within-checkpoint—one set of weights, $\max\lvert\Delta\Phi\rvert = 4.2 \times 10^{-4}$ at the worst seed—so free of the separate-tuning confound of §3.7, though not of the evaluation-time swap the first row of this block carries; a per-group affine recalibration fitted in sample removes 86% of the *MAE* gap and refitted out of sample removes none of it, and leaves this one untouched either way where the fitted map increases, as it does in 589 of the 823 groups in both arms at all three seeds. The substitution falls on the solvent channel
Closure against inputs on the activity axis (§3.5); margins in squared $\ln\gamma_2^\infty$ units, not the $\ln x_2$ MAE of the block above		
$B_{\mathrm{closure}} > B_{\mathrm{insuff}}$ on *glycol-ether* solvents[d,g]	broad IDAC, 182, 3 sources	the margin, a lower bound on $B_{\mathrm{closure}} - B_{\mathrm{insuff}}$, runs $+2.04$ $[+1.27, +2.83]$ to $+2.94$ over the four cells, $P > 0.95$ in all of them; worst deletion $+1.60$—and the survivor of a 59-fold search, at the ninetieth percentile of a chemistry-blind null that certifies some row set in 58% of draws, and not testable out of sample, where no rows of that cell exist
… on every other stratum of the map	broad IDAC, 59 strata	§S3.2, Tables S10–S14, each with its margin, interval and the test it fails; none is stated as a finding[h]
… for either set *as a whole*	broad IDAC (477); VT-2005-matched (60)	$+0.51$ (full $+0.95$); -0.38 compounding two cuts ($+0.01$ full); $+0.18$ less one publication; $+0.05$ on the smaller set: an average weighted by composition[f]
$B^{\mathrm{up}}_{\mathrm{insuff}}$ near stratum-independent, MSE not	broad IDAC, 3 boundable classes; 6 with an estimate	$\times 2.0$ against $\times 25$; $\times 6.8$ against $\times 51$; p=0.02 against 2000 null partitions of 9 classes
$B_{\mathrm{closure}} > B_{\mathrm{insuff}}$, 2010 on its *own* typed latent[f]	broad IDAC (477); VT-2005-matched (60)	$+1.29$, $P > 0.95/{\approx}0.9/{\approx}0.9$[e]; -0.77 at $P \approx 0.3$ on the smaller set, where the ordering is lost: loss of resolution, not a reversal
Whether a better closure moves the ceiling (§3.5.3)		
The 2002 $\to$ 2010 step lowers the error	broad IDAC (477)	-0.99 at $P \approx 0.2$: no improvement, and not a measured equality either; under the full convention -0.07, $P \approx 0.6$, and $+0.06$, $P \approx 0.6$ with dispersion on
…but does on the VT-2005 set's chemistry	57 shared pairs, UD	$1.78 \to 0.77$, $P > 0.95$, at fixed profile database and temperature, full convention; those 57 pairs are also broad-set rows, so not an independent sample
A fitted kernel residual transfers across solvents	VT-2005-matched (60), grouped folds	$-5 \pm 16\%$: spans zero; the $\pm$ prices optimiser noise only, and the fold spread that dominates it is not propagated, so no bound follows
Mechanism (§3.3)		
The latent departs from its reference along a shared direction	matched molecules (44), 3 seeds	$51 \pm 2\%$; the magnitude is not stable across analysis configurations, and these are a separate three-seed set of runs, not the arms of Fig. 2, whose latent departure is not measured
…and that departure's top-two mode share exceeds a phase-randomised null	one grounded checkpoint, same 44	0.40 against 0.22 ± 0.01, the $\pm$ being the null's spread; one checkpoint, and a wrong-molecule null is not cleared
…and that direction *transfers* to molecules outside the fit	matched molecules (44), 3 seeds	$4.1 \pm 0.5\times$ a zero-drift null, with no constant-offset control on these seeds; the $\pm$ spans seeds at one partition
That departure *cancels* closure error	not measured	the one direct check is under-powered and runs against it
Accuracy (§3.7)		
Physics arm vs. DirectGNN accuracy (ΔMAE)	labelled test rows (5608 of 8103)	$+0.14$ between separately tuned pipelines, so not attributable to the thermodynamic bottleneck

[a] The sets are defined in Table 1. [b] P is a two-way (solute $\times$ solvent) cluster-bootstrap frequency that the margin stays positive, reported as a band by the rounding rule at the head of this section. [c] The same 5608 rows: the seed-42 control arms carry the identical labelled set, so their cross-arm intersection with the arms of Table S2 is that set exactly. [d] One estimator cell (§2.5) for both sets, both kernels and every stratum. [e] Bootstrap band over pair / two-way / source clusters. [f] An aggregate margin is a composition-weighted average, and the deletions and cuts recorded here are failures of the headline cell rather than robustness passes. These rows record what the aggregate was and how it fails; the rows above replace it. [g] The comparison is one-sided in both directions it can be read: a positive margin establishes $B_{\mathrm{closure}} > B_{\mathrm{insuff}}$, a non-positive one is failure to separate and licenses nothing. The margin's size is a lower bound on $B_{\mathrm{closure}} - B_{\mathrm{insuff}}$ and not an estimate of it, so an interval on a margin bounds that bound and not

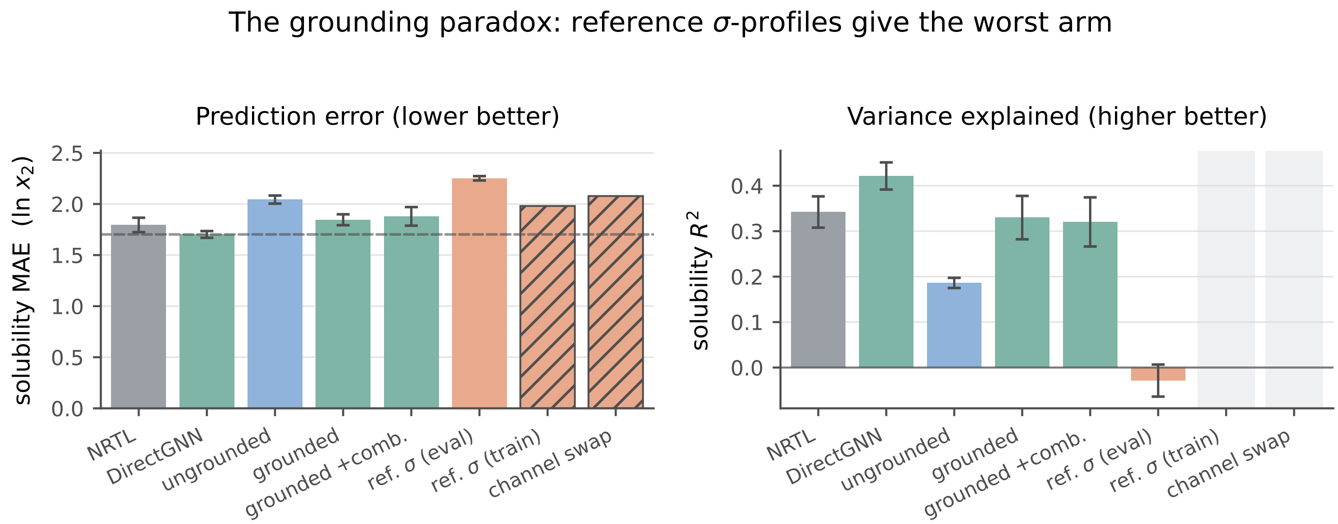

Figure 2: Controlled comparison on the 5608 labelled test rows (Table 1). Solid bars: three seeds, mean ± a population standard deviation (`ddof`= 0). The two hatched bars are the co-adaptation controls, seed 42 only, so they carry no error bar and no R^2, and keep an empty slot in the right panel. Dashed in the left panel only, the DirectGNN reference.

the grounded closure's by 0.013, so whatever produces it lives in the encoder and the data. Second, the reference-profile arm is the *least* shrunk and the worst. Its slope is the closest of the four to 1 and its prediction spread exceeds the measurement's, yet its MAE is the highest and its R^2 is not distinguishable from zero; substituting the reference restores spread that is not aligned with the truth, and its worst region is the sparingly-soluble tail it appears to reach. That is the misspecified map made visible on the solubility axis.

The substitution also costs the decision the model is used for, and here the separate-tuning confound of §3.7 does not stand in the way: the reference-profile arm re-runs the grounded checkpoint's own weights over the same rows and changes only the profile fed to the closure, so the two sides differ in Φ by 4.2×10^{-4} at the worst seed and no second tuning stands between them. The confound it does not remove is co-adaptation: like the headline MAE this is an input swapped at evaluation, no arm that had trained on the reference profile was ranked, and what co-adaptation would do to an ordering is unmeasured here. Grouping the labelled test rows by solute and temperature and ranking the solvents inside each group (823 groups over 107 solutes; Table S17), the mean per-solute Spearman correlation falls from 0.592/0.546/0.598 to 0.365/0.369/0.410 across the three seeds—pooled -0.197 $[-0.282, -0.117]$, every per-seed interval excluding zero, and every one of the three larger than any difference between seed replicates of a single arm, the largest of which is 0.060. nDCG@3 falls by -0.084 $[-0.124, -0.043]$. The solvent the model would select changes in 40 to 53% of groups, asymmetrically: the learned profile alone picks the measured best solvent in 20.0/23.2/21.5% of groups, the reference profile alone in 6.2/14.5/12.2%. None of that is a level. Fitting a separate affine recalibration inside every group—a slope as well as an intercept, the largest concession a reading on which the substitution merely miscalibrates can ask for—removes 86% of the MAE gap, $+0.431$ $[+0.308, +0.553]$ to $+0.060$ $[+0.021, +0.101]$, which is the restored, misaligned spread of the paragraph above in another currency. That map is fitted on the rows it is then scored on, two free parameters against a median of five solvents, so it bounds what the concession can buy rather than measuring it: refitted leaving out the row it scores, it removes none of the gap, $+1.09$ $[+0.3, +2.2]$ pooled, and on the 550 groups holding at least five solvents, where a refit has something to work with, +0.28/+0.07/+0.04 by seed with two of the three intervals spanning zero (§S6.1). The MAE penalty is level and scale in the sense that a per-group map fitted on the answer absorbs most of it; out of sample nothing that sharp can be said in either direction. The ranking penalty is not, under either fit, and by invariance rather than by any centring argument—though the invariance is to an *increasing* affine map, and the fitted map decreases in 10 to 12% of learned-profile groups and 17 to 19% of reference-profile ones, where it reverses that group's order outright. Restricted to the 589 groups of 823 whose map increases in both arms at all three seeds, the ranking gap is ρ -0.148 $[-0.218, -0.079]$ and nDCG@3 -0.068 $[-0.110, -0.026]$, with top-1 then spanning zero.

What stands in its place cannot be told apart from a solute-blind ordering. Set each arm against two references that know nothing about the solute—the same arm's own predictions for a randomly drawn other solute over the same solvents, and a ranking of those solvents by their leave-one-solute-out mean measured solubility—and the learned-profile arm is above both on Spearman, Kendall, top-1 and nDCG@3 at all three seeds, twenty-four intervals of twenty-four excluding zero. The reference-profile arm's own twenty-four margins are small and all but one positive, and it is their intervals that carry the reading: twenty-four of the twenty-four include zero, or twenty-three, one cell sitting on the boundary at $[-0.0008, +0.0762]$ and changing side with the bootstrap draw (Table S17). What that measures is a failure to separate the reference-profile arm from orderings built without the solute, not a demonstration that it cannot be separated from them. Neither arm is chance, the within-group permutation floor being $\rho = 0.00 \pm 0.02$ for both. Supplying the true physical intermediate does not leave the model miscalibrated about solutes; it leaves its ordering of them indistinguishable from one drawn without them.

3.3 What moved inside the model, and whether a downstream correction repairs it

A candidate *mechanism* for the low-fidelity, end-to-end regime is this: a σ-profile head trained *end-to-end* through a fixed, misspecified closure—with no σ supervision in that phase—is under pressure to *cancel* the closure's error rather than to be physically faithful. The hypothesis has two parts and this section establishes only the first. Its antecedent—that the latent leaves the reference along a direction shared across molecules rather than as per-molecule noise—is measured. Its consequent—that the departure cancels part of B_{closure}—is not, and the one direct check available argues against it. The external reference that fixes the ceiling makes the first part testable: we compare the predicted profile $\hat{\sigma}$ against the reference VT-2005 profile molecule by molecule (n=44). That comparison is the anatomy of the object the two operations of §3.1 act on—how far the learned intermediate has travelled from the reference that supervises it and then replaces it—and it is placed here, inside the phenomenon, for that reason and not as evidence for why either operation has the sign it does.

The drift under fine-tuning. The comparison is within a single grounded model, checkpointed before and after solubility fine-tuning (the σ head unfrozen), so architecture and initialisation are held fixed and only the objective differs; it is repeated over *three seeds* and reported mean±sd. Those three runs are not the arms that produce the paradox: they are a separate set, trained with the σ head left unfrozen through fine-tuning, where the four COSMO-SAC arms of §3.1 hold the head from P2 onward (Table S2). The freeze holds the head and not the encoder feeding it, so those arms' profiles are not pinned at their warm-up value either; how far *their* latent drifted is not measured, and the deposited per-arm predictions carry no σ-profile, so it cannot be recovered from them. What follows is therefore measured on runs of its own and is not estab-

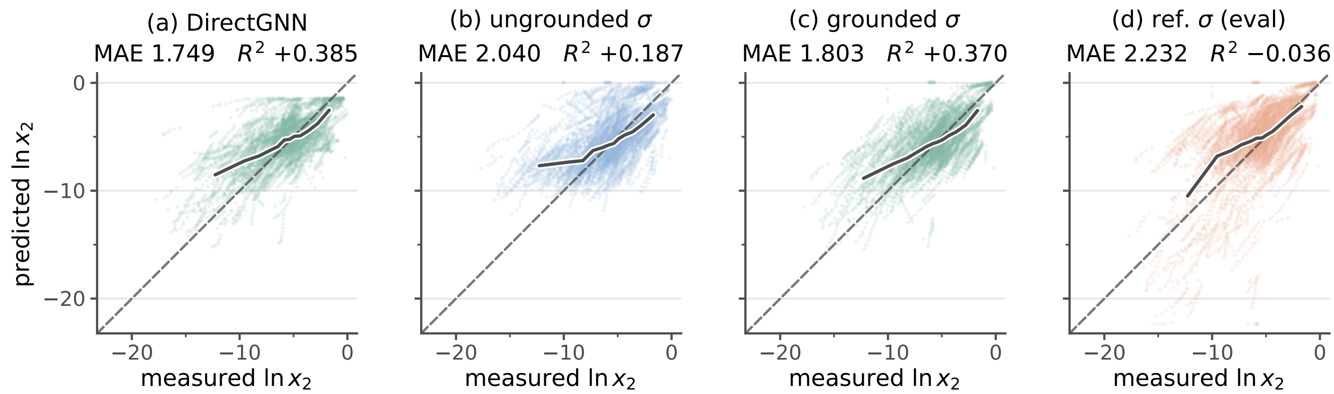

Figure 3: Solubility parity on the scaffold-split test rows: predicted against measured $\ln x_2$ for four arms of the controlled comparison. All four panels are the same 5608 labelled test rows (Table 1), from seed 42 of three. Those rows are 763 solute–solvent pairs over 147 solutes and 70 solvents, measured between 248 and 368 K, so one pair contributes a short temperature series and the number of points exceeds the number of independent systems. (a) is the closure-free control; (b) is COSMO-SAC over an unsupervised σ-profile, (c) the same closure with σ-profile supervision during training, and (d) the arm of (c) with the reference VT-2005 profile substituted at evaluation. Dashed, the identity; solid, the median prediction in twelve equal-count bins of the measurement. The MAE and R^2 above each panel are that panel's own, on those rows and that seed, to the three decimals Table S2 uses for a per-seed value; the three-seed means are Fig. 2 and that table, and differ from these by at most 0.05 in MAE. Only (d) carries scatter below the identity that reaches $\ln x_2$ values no measurement occupies. The axes span the union of measured and predicted values over all four arms; no point is clipped.

lished of the checkpoints whose paradox it is offered to explain. Under σ supervision the head reproduces the reference profile to within 36±1%, so the intermediate *can* carry the physical shape, though only as an in-sample fit: all 44 probe molecules lie inside the σ-grounding pool. Solubility fine-tuning then drives the profile 41 ± 3% off the grounded one, for a total departure from the reference of 51 ± 2%. The drift is not per-molecule noise, and its structure is not a constant offset: the first two principal components of the *mean-centred* deviation carry 72 ± 1% of the variance. The quantity read against a null is a different one, the top-two share of the learned-versus-reference deviation, 0.399; its null is *phase-randomised*, matching each deviation's own smoothness and destroying only the between-molecule alignment at issue, and the observed share clears it (0.399 against 0.221, $p < 0.001$), at roughly 1.8×. The 72±1% mean-centred share has no matched null of its own, a wrong-molecule null built from real reference profiles does *not* clear (0.772, p = 1.00), and if the case for preferring the phase-randomised null is not accepted the spectrum is uninformative rather than supporting. To test whether the structure transfers across molecules we fit one distortion operator D on half the 44 molecules and score it on the other half, where it predicts the drift 4.1±0.5× better than the zero-drift (identity) null out of sample. That null is deliberately weak, nothing controls it on these three seeds, whose checkpoints were not retained, and a molecule-independent constant offset reaches 2.1× on the one companion run where that control can be computed, so the ratio stands against the identity null alone; the held-out half is homologue-saturated. Structure *beyond* an offset therefore rests on the mean-centred spectrum, not on the ratio—and that spectrum does not exclude a shrinkage of every profile toward the corpus-mean shape. On the reference ladder of §S5 the drift consumes essentially all the molecular specificity grounding had provided, as such a collapse would also look.

Dominant mode, and the direct cancellation test. The dominant drift mode (Fig. S4) transfers area *out of* the nonpolar bins ($\sigma \approx 0$) and *into* the polar wings: the network represents molecules as more polar than they are. Whether that shift *cancels* part of B_{closure}—the input compensating the map, as the non-identifiability of the split allows (Lemma S1)—is the comparison $\mathrm{MSE}(m, g(\hat{\sigma}))$ against $\mathrm{MSE}(m, g(z^{\star}))$ on the same pairs. It does not support the cancellation reading. Holding the cavity area at its reference value so that only the profile shape varies, the sole retained end-to-end SLE-trained COSMO-SAC checkpoint gives 18.75 against 1.47: on the activity axis its drifted profile is an order of magnitude *worse* through the same closure, not better. Three limits sit on that number. The checkpoint is the uncontrolled one, whose departure (82%) is half again the three-seed 51 ± 2%. The axis is wrong: the drift is produced by the finite-composition solubility objective and scored here at infinite dilution. And the check has almost no power in the direction that matters, so it separates "no cancellation" from "large cancellation" but not from "some cancellation". It is the only direct evidence available, and it is evidence against rather than a clean pass or fail. A second, independent check on the closure's own axis points the same way: a *more* polar profile moves g further from m (§S5). Both sit on that same infinite-dilution axis, so neither refutes cancellation in the regime that generates the drift; they remove the one place it could have been measured, and the cancellation reading remains inferred from the paradox itself. What is established is weaker and still consequential: the model's "physical" intermediate is not the molecule's charge distribution, consistent with cautions that the interpretability of physics-grounded latents is not guaranteed under misspecification,[20,29] and that adapting a representation can itself collapse an otherwise physically-faithful latent.[30] Its *structure*—low-rank rather than diffuse—excludes a random collapse but not the structured shrinkage above.

What leaves σ under-determined by activity, and what freezing it buys. The freedom that would make such a cancellation possible is the closure's geometry, measured in §S5: at infinite dilution the activity map concentrates on one or two hydrogen-bonding directions, so the end-to-end head can move σ along the remaining, low-sensitivity ones at little cost to the fit—the σ-profile form of Lemma S1. Nothing there shows recovery of the physical profile to be impossible in principle. The sign of grounding turns on *both* axes, and on which sense of grounding is meant. Freezing the σ head at its supervised warm-up state instead of letting it drift end to end cuts the reference-input penalty by 65% (ΔMAE from +0.40 to +0.14 over n=5571 matched test rows). Those are a separate matched pair of runs, one base resumed twice with the freeze off and on, and not the +0.41 headline arms of §3.1. A near-physical intermediate makes the pipeline far more robust to substituting the reference σ. Supervision alone does not, however, *flip* the sign here—the penalty stays positive, the closure still being the low-fidelity COSMO-SAC-2002 and the warm-up σ itself only partly physical. The full flip needs supervision *and* fidelity together, which is what TeNNet-SAC[7] has and this pipeline does not.

Whether a downstream correction repairs it. If the closure binding were merely an inaccurate point estimate, a learned output-space correction on top of the physics prediction should recover accuracy. It does not: an additive bounded residual with a learned confidence gate leaves the gate effectively shut, and forcing it open removes the systematic *bias* (+0.45 $\to$ +0.06) but recovers little *accuracy* (R^2 0.236 $\to$ 0.274 on the same rows), as expected if the closure, not the estimate, is the binding constraint. Both arms are seed 42 only, and this is a statement about the one correction tested: an additive bounded residual on the output, gated, on this closure and this corpus.

3.4 What the substitution does not decide, and what would settle it

Which of the two channels the substitution fails through is not decided here. Drug-like solutes are essentially absent from the VT-2005 set, so the substitution falls almost entirely on the *solvent* channel, and the both-channel substitution the attribution would need covers only ≈7 solutes; enlarging that set would mean computing new reference profiles rather than looking them up.

Nor does anything above decide *why* the substitution hurts. The reading it invites is that handing the closure a faithful input surfaces a misspecified map rather than repairing it, so that the reference profile is the better input only where the map that consumes it is faithful. This paper states that reading and does not assert it, and the reason is stronger than a test that returned nothing: on the solubility axis the two terms of the split do not separate at all, and the solubility axis is where the substitution of §3.1 was measured. The quantities the reading is about—a fixed map's own error, and what its inputs fail to carry—are therefore not observable on the rows §3.1–§3.3 report.

What follows in §3.5 is a second axis and not an explanation of the first. It is a different reference database, a different error functional, no trained model at all, and a row set that shares none of the rows above; the two are measured in different regimes on largely disjoint chemistry, and no measurement in this paper connects them. Until the connecting experiment is run, "the substitution hurts *because* the map is misspecified" is an inference and this paper does not assert it.

Two experiments would settle it, and both are stated as limitations rather than as prospects. The first is the transfer back to the paradox's own regime—finite composition and all γ, where the substitution is measured, rather than infinite dilution, where the split is—which needs a set matched across the two reference databases on which both can be computed on the same rows; that is limitation (i) of §3.7 and it is the principal open question here. The second is a typed σ-profile head trained end to end through a modern kernel, which is what would show whether the substitution's sign survives a closure of different fidelity: limitation (x). Neither has been run.

3.5 A second axis: bounding a deployed COSMO-SAC against its inputs

The estimator is the one cell fixed in advance in §2.5 and used nowhere else in this paper: eight equal-count bins of the conditioning variable, the residual convention, the within-bin Bessel correction, and the admissibility clauses (a)–(c) that decide which strata may be read at all. All four were fixed before any margin was computed; none of §3.1–§3.4 is measured with them.

The decomposition of §2.1.1 separates a fixed closure's error into a part in the map and a part in its inputs. Its output is a map over chemical strata rather than one verdict for a set, for the reason §3.5.1 establishes, and it runs on two data sets that must not be conflated, the *broad IDAC set* and the *VT-2005-matched set* of Table 1.

3.5.1 The broad IDAC set (n=477)

The closure reads a temperature as well as a profile pair and is evaluated at each row's own temperature, so the conditioning variable is its full argument $(z^{\star}, T)$—the concatenated UD profile pair with T, 103 dimensions against n=477 ($\dim/n \approx 0.22$)—and the target m is the experimental $\ln\gamma_2^{\infty}$, so floor and closure error are measured on one data set and one profile database. Its chemistry is small: no solute exceeds 12 heavy atoms (median 7), against a median of 14 and a maximum of 122 for the drug-like solids of this paper's own corpus, and only 0.46% of corpus rows carry a solute from this set. *Broad* therefore names the reach of the γ^{∞} corpus the kernel is calibrated on and not coverage of the chemistry a solubility model is deployed against: it covers the solvent side of deployment and not the solute side. It is matched to UD by InChIKey with a first-block fallback, looser than the VT-2005 set's rule; a wrong $z^\star$ inflates B_{closure}, so the looser rule runs against the ordering we claim, and the aggregate margin is unchanged on the 465 exactly matched rows.

Every MSE, bound and margin on this axis is in squared $\ln \gamma_2^\infty$ units, and none of them is on the $\ln x_2$ axis the MAEs of §3.1 are measured on. Under the convention rule of §2.5 the fair 2002 arm scores MSE = 1.902 against $\mathrm{Var}(m) = 4.85$, the LOTV bound gives $B_{\text{insuff}} \lesssim 0.70$ and hence $B_{\text{closure}} \gtrsim 1.21$: a separation margin of +0.51 for the set taken whole, +0.95 under the full convention and +0.27 collapsed to one row per distinct pair, positive in every cell of the bin-count × variance × convention grid, with two of the three bootstrap clusterings extending just below zero.

That aggregate is not the instrument's output. Two operations reported separately—the pair unit, and dropping the two chemistries the error concentrates in, the glycol-ether solvents and the rows carrying water as the solute—compound badly. The chemistry cut alone takes +0.51 to −0.12 on the 258 rows that remain, the pair unit alone gives +0.27, and the two together give −0.38 on 136 pairs; under the full convention the same pair of cuts takes +0.95 to +0.28 to +0.01. Nor is the aggregate carried by the set: deleting one source publication at a time, `10.1021/acs.jced.7b00114`—37 rows, six molecule pairs, 32% of the set's squared error—takes the margin to +0.18, while each of the other fourteen deletions leaves it between +0.42 and +0.87 (under the full convention that curve is flat, [+0.82, +1.07] against +0.95). That single deletion loses two thirds of the margin and its resolution without reversing its sign; the sign reverses only under the compounded cut. Two caveats qualify the corroborating estimators and not the bound: under random folds they leak, and under folds blocked on molecules they are uninformative on the VT-2005-matched set and split on the broad set; and the binning-free pair bound, tighter though it is, is not promoted, because its multi-row pairs each come from a *single* publication and it therefore samples no between-laboratory component. Together they say the binning bound is slack rather than that B_{insuff} is negligible. What is reported rests on the LOTV bound (§2.5, Table S8).

The map, and how much of it survives the rule. The instrument runs over 59 strata of the broad IDAC set—477 rows, 185 pairs, 78 solutes, 36 solvents, 15 source publications—on the solvent axis at three granularities and the solute axis at two, plus their cross-tab and the set as a whole. Fifteen have $n \geq 40$ and are boundable, the only ones whose margins the rule can act on; the other 44, including the 24 that carry no estimate at all, are printed with the test each fails in §S3.2 (Tables S10–S14 and Fig. S2), every stratum with its error, its bound, its interval and its share of the set's squared error, and nothing dropped for being small. Six of the fifteen are *admissible.* Clause (c) of the rule makes those six three distinct row sets, of which one is both positive and not demoted by its residual test: the glycol-ether solvents. **That row set is the one the admissibility rule leaves standing**, at a margin a chemistry-blind relabelling of the same molecules reaches in 10.1% of draws (below): on 182 rows over 43 pairs from three publications carrying 45% of the set's squared error, the closure's own error exceeds what input insufficiency can account for by at least +2.04, that bound carrying a 90% interval of [+1.27, +2.83] at the deployed convention—+2.94 at the pair unit under the full one—positive in all four cells and under every single-publication deletion, the worst leaving +1.60 when `10.1016/j.jct.2016.10.013`, which carries 53% of that stratum's squared error, is removed. It is not an artifact of one solute family: the 74 rows left when its alkane solutes are dropped keep the sign at +1.40 deployed and +1.69 full, under all three deletions. Nor is it a property of the estimator cell. Swept over the bin-count grid Table S4 runs for the aggregate—the analyst choice with the largest leverage here, since that grid takes the aggregate from +0.01 to +0.96—the stratum is positive in all 56 row-unit cells and all 40 the pair unit defines, the deployed unbiased column running +1.06 to +2.19, and the deletion test passes at every one of the 36 cells where it can be run (Table S5). The occupancy rule that fixed the eight-bin cell would ask for 23 bins at n=182, so the cell the map reads is the coarser and the lower of the two. That sweep varies the coarsening's own settings and not the family of estimator: a pre-declared cross-fitted regressor, tested after this analysis was fixed, returns a *looser* bound and moves this stratum to +1.76, leaving it the one admissible positive row set but with fewer cells around it (§S3.3). What the sweep establishes is therefore robustness within the coarsening, not across estimators.

What the row set does not have is a multiplicity null it clears, and this belongs beside the finding rather than in a limitation. It is by construction the survivor of a 59-fold search. Under 2000 chemically blind relabellings of the same molecules—each solvent and solute keeping its rows and taking another's class labels, the stratum family rebuilt and put through the same rule—the map's six admissible strata are above chance ($p = 0.01$) but its two admissible positive row sets are not ($p = 0.29$): such a relabelling certifies at least one row set in 58% of draws. The size is the sharper statistic, and it is not sharp enough: a certified margin of +2.04 or more arises in 10.1% of all draws, which is 17% of the draws that certify anything, against a median of +1.57 among those. The separation is what the instrument licenses on the rows it was computed on; it is not established as a chemistry-specific effect against the search that found it. Each p here is a frequency and not a bound in either direction: the null's own two biases are measured rather than argued and run against each other (§S3).

What settles a selection objection is the same measurement on rows the search never saw, so we ran one. The out-of-sample set is the PGL 6th-edition IDAC database scored by the same deployed closure on VT-2005 profiles (Table 2): another group's compilation, sixteen times the broad set's rows on a different profile database. The row set, the estimator cell, the admissibility rule and what would count as confirmation in each direction were written down before any margin was computed (§S3). **The chemistry could not be tested there**, for a reason about the data rather than about the number: the cell the finding is about—a glycol-ether solvent with an *organic* solute—has no rows there. The non-aqueous half of that database carries 88 solvent molecules and not one glycol ether. *Not testable* is the third verdict the pre-declaration lists and counts as non-confirmation, and it follows from the row inventory alone: what the pre-registration delivers here is the absence of confirming data and not a failed replication. What that database does carry is 52 rows of water in seven glycol-ether and polyol solvents over seven pairs, drawn 50–2 from the two compilations of one record file; deleting the larger of the two leaves two rows and the fixed cell undefined, so the sign is never put at risk and the row set is inadmissible under the same rule that governs the map—the rule refuses it out of sample for the reason it refuses water-as-solute in sample. Its margin, which therefore licenses nothing, is +0.35 [+0.28, +0.58]. Only ethylene glycol is shared with the four solvents of the in-sample cell; the three oligomers carrying 141 of its 182 rows are absent. What belongs with that verdict is the whole out-of-sample ranking rather than the part of it that runs this closure's way, stated as description and not as a test, and that ranking is Table S9: over the database's ten solvent classes the same cell admits two, the mono-alcohols—whose margin is positive with an interval covering zero—and water, whose margin is negative. No out-of-sample class both passes the rule and separates the two terms, and every margin above the mono-alcohols' sits on a class the rule refuses. Out of sample the glycol ethers are neither where this closure fails hardest nor a cell the rule can read. The chemical localisation is therefore what the instrument licenses on the rows it was computed on, and the title does not carry it.

It is also the whole of the separation this paper reports, and the statement carries five boundaries that the phrase "COSMO-SAC error exceeds input error" does not: no other stratum of the map returns one (below); neither data set taken whole does (the aggregate is retired above, and the VT-2005-matched set's +0.05 does not stand, §3.5.2); nothing on the solubility axis does, where the two terms do not separate at all and where the substitution of §3.1 was measured; the error that stands above the bound is COSMO-SAC's *as deployed*, the composite of map and computed conformer that the pipeline runs on rather than the kernel (§3.5.3); and the margin is one its own chemistry-blind null returns at the ninetieth percentile, on the one chemistry no out-of-sample rows exist for.

Which chemistry does not separate. Three of the six admissible cells are that same row set at three granularities. The remaining three are two row sets and neither establishes anything: the alkane solutes are admissible and positive and clause (c) demotes them to the glycol ethers, which hold more rows and hold most of theirs, while the acceptor-only aprotics pass the rule and license nothing, the instrument being one-sided. That is the whole of the uniqueness claimed for the glycol ethers: the only row set admissible, positive and left standing by clause (c)—*not* the only admissible stratum and not the only one sign-stable under every deletion. Which cell each of the remaining strata fails on, and by how much—too few rows to bound, one publication so that the deletion cannot be run, or a sign lost under a deletion—is in §S3.2, and none may be used to explain a reversal. The whole-set row passes at the headline cell and fails once the pair unit is required, so the rule does not by itself dispose of the aggregate there; the composition argument above does.

Why the output is a map. $B^{\mathrm{up}}_{\mathrm{insuff}}$ is nearly stratum-independent and MSE is not. Across the three classes the design can bound—glycol ether, water and the acceptor-only aprotics, bounds 0.108, 0.215 and 0.140—the bound spans a factor of 2.0 against errors (2.252, 0.476, 0.089) spanning a factor of 25; adding the three that carry an estimate without being boundable—mono-alcohol 0.733, halogenated 0.592, N–H protic 0.281—gives 6.8 against 51, that 6.8 being set by the mono-alcohols' 0.733, which exceeds that stratum's own $\mathrm{Var}(m) = 0.604$ and is flagged in the artifact as an invalid bound on a stratum already excluded. The two orderings are unrelated—among those six the glycol ethers carry the lowest bound and the second-highest error—and against 2000 random partitions into blocks of the observed class sizes, at the same cell, the chemical partition's range (largest minus smallest) in MSE exceeds every draw while its range in the bound is unremarkable ($p = 0.26$); their ratio gives $p = 0.02$. That is the estimator showing through: every stratum is summarised by eight quantile bins of one scalar, so the bound measures the residual spread of m inside such a bin, a property of the binning rather than of the chemistry, while nothing constrains the error to follow it. An aggregate margin is therefore a composition-weighted average of a term free to move 25-fold offset against one that barely moves, and it will report a binding closure for any set whose composition happens to include a chemistry the closure fails on—a statement about the aggregate, not about the bound, which stays valid in each stratum where it is computed.

3.5.2 The VT-2005-matched set (n=60)

This set carries the paper's other reference-matched analyses, and it is far weaker. All 60 rows sit at 298 K, so $z^\star$ is the profile pair alone and $\dim(z^\star)/n \approx 102/60$ leaves the split behind one-sided bounds; it is strongly unbalanced, methanol occupying 25 of the 60 rows and the two worst-fit pairs carrying 34% of the squared error; and its 17 solvents fall in six of the nine classes, none reaching the n=40 a bound needs, so it supports no stratified bound and no stratum of it is stated as a finding. Its aggregate margin is +0.05, and it does not stand: deleting the two leverage pairs takes it to −0.002, dropping methanol loses it, blocked out-of-fold estimators do not support it, and 11 of the 60 single-pair deletions take it non-positive, where on the broad set no single-pair deletion moves the sign at all. Zero-mean label noise can understate a separation but not create it (§2.1.1); what a margin does bound is a label systematic tracking $z^\star$, implausible on the maximum-likelihood binning and *not excluded* on the Bessel-corrected one. Of the 60 pairs, 57 also appear in the broad IDAC set at 298 K under UD profiles, so the two are not independent samples—this is the sub-design VT-2005, and therefore the paradox's own oracle, is defined on, and the agreement between the two sets' error orderings is a consistency check rather than corroboration.

3.5.3 Whether a better closure moves the ceiling, and where this one fails

If the ceiling is the closure, the direct remedy is a better closure. The standard 2002 $\rightarrow$ 2010 step delivers one on the VT-2005-matched set and not on the broad IDAC set, and a block split identifies the factor that carries that reversal. The step more than halves the smaller set's error under the full convention ($P > 0.95$), the convention under which the two kernels' published error figures are reported, and cuts it 42% under the deployed one; on the broad IDAC set it is not an improvement under either rule. The reversal is the *chemistry* and neither the profile database nor the temperature range—with UD profiles, one extraction and one convention throughout, the *2010, dispersion off* arm wins on the 57 pairs shared with the smaller set and does not win on the complementary 420—and that block split is the whole of the evidence for the reversal being chemical. We claim only that **the fidelity lever we can demonstrate is confined to the smaller set's chemistry**, and that null is additionally a small-molecule and glycol-heavy one, so on drug-like solids the lever is untested rather than absent. The input-fixed control and its three limits, the three arms on both sets under both conventions (Table S7), the third block at 298 K, and the floor checks under the 2010 kernel's own typed latent—where the ordering is lost on the smaller set and holds on the broad one, loss of resolution rather than a demonstrated flip to the inputs—are in §S6.6.

Where B_{closure} dominates—on this set, the one row set the rule leaves standing—the ceiling is in the map *as it is deployed*: the fixed kernel together with the one computed conformer it reads, which is the composite B_{closure} is the error of (§2.1.1) and the object the pipeline runs on. Which of the two carries the error is a question this section does not answer; nor does dominance say the composite is *fixably* wrong. A small structure-preserving residual on the fixed COSMO-SAC exchange-energy kernel, fit on the reference VT-2005 profiles, removes $46 \pm 5\%$ of the reference-input error under random folds but $-5 \pm 16\%$ once folds are grouped so that no solvent is shared. Only the grouped design is free of leakage, the closure error being predominantly a between-solvent systematic. It gives a null and not a bound: the $\pm$ prices only optimiser noise over five fit seeds, the held-out fold spread that dominates it going unpropagated (§S3). B_{closure} is therefore an error a handful of kernel parameters can absorb *within* a solvent and that we have not shown to be correctable across solvents—a statement about what a correction of that shape can absorb, not about which component of the composite the error came from.

Where it fails is chemically placed, and placed differently on the two sets. On the VT-2005-matched set 90% of the squared error falls on strong hydrogen-bond-donor solvents and the closure is near-exact on acceptor-only ones—the chemistry COSMO-SAC's documented single-parameter H-bond term governs, the term the donor/acceptor-typed 2010 closure was built to replace. Half of that reading carries to the published evaluation of Table 2 and half does not, and on the broad set neither it nor its replacement reproduces, so we state neither (§S6.6). It places the error in a *chemistry* and not in one component of the composite: the reference profile for each of those solvents is one gas-phase conformer too, and neither set separates a displaced reference from the kernel.

That COSMO-SAC-2002 is a low-fidelity closure is the contribution rather than a qualification of the result: it is what a practitioner adopts by default, a closure's fidelity is *not knowable in advance* on a new system, and the field routinely composes an expressive encoder with a fixed mechanistic map and assumes the map is good enough. What follows when it is not is that the learned latent silently departs from the quantity it is named after (§3.3), invisibly without an external oracle. Outside that one row set the map reports cells it cannot resolve, not cells in which the closure is sound.

One arithmetic sizes that exposure. The ground truth charging B_{closure} is quantum chemistry and not experiment: the VT-2005 profiles are computed for a single reference conformer and protonation state, and the successor Delaware database revises those conformations for about a third of its compounds;[31] where the reference is displaced, even a perfectly specified g carries a term in B_{closure} through its curvature (§S9). Table 2 measures that axis directly—one closure, two profile databases, a factor of 2.6 in AAD, 6.6 in squared error if the errors scale uniformly—while the glycol-ether stratum's margin closes only once its squared error falls below twice its 0.108 bound, a more than ten-fold reduction (Table S14). A profile source better than VT-2005 by the whole of that published bracket would leave that sign standing. The arithmetic holds the insufficiency bound fixed, which a new profile source would not, so it sizes the exposure and does not bound the share.

3.6 The same split on a second closure: pKa through a fixed Hammett relation

The instrument is not specific to COSMO-SAC, and a second domain runs it where the sign of grounding can be predicted before the measurement. The closure is a Hammett linear free-energy relationship $g(z^\star) = \mathrm{p}K_{a,0} - \rho\,\sigma^\star_{\mathrm{H}}$ in a substituent constant, with $\mathrm{p}K_{a,0}$ and ρ the tabulated Hansch–Leo–Taft reaction-series constants, one literature pair per scaffold—$(4.20, 1.00)$ benzoic acid, $(9.99, 2.23)$ phenol, $(4.60, 2.77)$ anilinium, $(5.25, 5.90)$ pyridinium—never fitted or refit on these data, so g is fixed in the sense Lemma 1 requires and the same g runs inside every split. It reads two things, so $z^\star$ is the pair $(s, \sigma^\star_{\mathrm{H}})$: the series s, which selects that literature pair, and the substituent constant at the tabulated value the rule below assigns. Only the constant is learned—the learned intermediate $\hat{\sigma}_{\mathrm{H}}$ is that constant predicted from structure; the external reference is the Hansch–Leo table,[32] printed in full in §S8; and the data are the OPERA experimental compilation[33] under a uniform, closure-independent quality filter that drops pre-ionized microstates.

The reference constant is assigned mechanically, and that rule is also what defines the two poles. The four kinds are tried in the fixed order benzoic acid, phenol, anilinium, pyridinium, and a record enters under the first of them whose ionizing group matches exactly once in the molecule, at a ring atom lying in exactly one ring and that ring six-membered and aromatic. An ionizing site the entering kind does not consume does not exclude the record: on the same ring it is typed as an ordinary substituent, tabulated or not. It enters only if its ring system is also unambiguous: no aromatic nitrogen anywhere in the molecule for the three carbon-ring series, exactly one and no N-oxide for pyridinium. The kind a record enters under names the reaction series and with it $(\mathrm{p}K_{a,0}, \rho)$. Ring positions are typed by shortest ring-bond distance from that ring atom—the one bearing the ionizing group, which for pyridinium is the ring nitrogen itself—one bond *ortho*, two *meta*, three *para*; and $\sigma^\star_{\mathrm{H}}$ is the sum over the substituted positions of the Hansch–Leo constant of that substituent *at that position*, σ_m meta and σ_p para, taken from the fixed list of nineteen substituents printed in §S8; an unsubstituted parent has $\sigma^\star_{\mathrm{H}} = 0$. An ortho position carries no tabulated constant and enters the sum as zero, and so does a substituent outside the list. The first pole—the *clean* one below—is the 158 records whose every substituted position is meta or para with a tabulated substituent; the *degraded* pole is the 846 where at least one position is neither—571 of them carrying an ortho substituent, the other 275 an untabulated one—and there $\sigma^\star_{\mathrm{H}}$ is the partial sum over the positions it can see. The split is not benzenoid against heteroaromatic: all four series occur on both sides, 32 of the clean pole's records being pyridinium.

The two classic failure modes of that relation fall onto the two channels of Lemma 1: *resonance saturation*, a curvature the linear closure cannot represent, lands in B_{closure}, while what the zeroed positions carry—ortho field and steric effects, and whatever an untabulated substituent contributes—is information the intermediate lacks and lands in B_{insuff}. The two poles differ accordingly. On the clean one, the chemistry the relation was built for (n=158), it is well specified: RMSE 0.55, and the lower bound MSE$-B^{\text{up}}_{\text{insuff}}$ on the non-negative B_{closure} is -0.2 (95% CI $[-0.4, -0.0]$)—below zero, where a lower bound sits when the closure predicts as well as a flexible regressor on its own argument—while the per-scaffold systematic offsets are ≤ 0.19 in pK_a, whose square is the independent Jensen lower bound: ≤ 0.034, 11% of the 0.303 total. On the degraded pole (n=846) it degrades to RMSE 2.26, of which the same estimator places at most 4.5 of the 5.1 total in input insufficiency and hence at least 0.6 in the closure—both figures one-sided, in the same direction as the solubility bounds.

That estimator is not the eight-bin coarsening cell of §2.5, and its conditioning variable is not the constant alone; nor is the functional the same, the two figures just given being floors MSE $-$ $B^{\text{up}}_{\text{insuff}}$ on B_{closure}, which when positive establish $B_{\text{closure}} > 0$, and not the separation margin MSE $-$ $2B^{\text{up}}_{\text{insuff}}$ of Table 1, which when positive would establish the stronger $B_{\text{closure}} > B_{\text{insuff}}$. The closure reads the reaction series too—p$K_{a,0}$ and ρ are per-series—so the argument Lemma 1 requires is the pair $z^\star = (s, \sigma^\star_{\text{H}})$, series label and constant, and the estimand is $\mathbb{E}[\text{Var}(\text{p}K_a \mid s, \sigma^\star_{\text{H}})]$. Neither coordinate needs binning: s takes four values and $\sigma^\star_{\text{H}}$ is a scalar, so the conditional variance is estimated directly—a random forest of pK_a on $\sigma^\star_{\text{H}}$ alone, fitted *within* each series and pooled over series by n, five-fold, its *out-of-fold* residual mean square. That regressor $\hat{r}$ reads $(s, \sigma^\star_{\text{H}})$ and nothing else, and predicts each row from the folds that exclude it, which is what makes the split exact for it: $\mathbb{E}[(m - \hat{r})^2] = \mathbb{E}[\text{Var}(m \mid s, \sigma^\star_{\text{H}})] + \mathbb{E}[(\mathbb{E}[m \mid s, \sigma^\star_{\text{H}}] - \hat{r})^2]$, so the estimate exceeds B_{insuff} by the regressor's own approximation error—an upper figure, leaving a lower one. Conditioning on s is not a free choice and it is not the conservative one: pool the four series and the same estimator returns 8.3 against the degraded pole's MSE of 5.1, a conditioning under which g is not measurable, so Lemma 1 does not license it and the closure floor it leaves would in any case be vacuous. The floor of 0.6 therefore rests on s belonging in $z^\star$, which is what the lemma demands of a closure that reads it. §3.5.1 declines fitted estimators on solubility because random folds let a held-out row borrow strength from a near-twin through the representation; here the regressor's only input is $\sigma^\star_{\text{H}}$ within a series, so no fold can hand it what the closure's argument does not already carry. What the identity needs of the fold design is only that a row be held out of the fit that predicts it.

The decisive test is the same oracle swap, measured *in-distribution* across K=20 within-scaffold splits: the learned $\hat{\sigma}_{\text{H}}$ replaced by the tabulated $\sigma^\star_{\text{H}}$ in the same closure—the series label is read from the structure in both arms, so the swap moves one coordinate of $z^\star$—the sign of the change recorded per stratum. It flips. On the clean pole the reference *improves* accuracy—MAE 0.48 learned against 0.315 with the reference, better in 19 of 20 splits and in three of the four reaction series—so the ceiling there is the noisy learned estimate and not the closure, which is an internal positive control on real data. On the degraded pole the same swap *degrades* it—0.83 against 1.62, worse in 20 of 20 splits and in all four series, gap 90% CI $[+0.57, +0.95]$—the freely learned constant absorbing what the zeroed positions carry and the tabulated one lacks. The sign of grounding therefore reverses inside one closure, in the direction Proposition 1 gives a priori and—consistently with the one-sided bound above, though not isolated by it—through the insufficiency channel rather than the closure-error cancellation of §3.3. Both signs are one comparison: the sign of the grounding gap is by construction that of the learned arm's error against the fidelity-set oracle error, and what a competence sweep adds is that the oracle side of it does not move with the training budget while the learned side does. The clean pole's threshold ($\approx$ 0.315) is never beaten by the learned arm, so grounding helps at all four budgets swept, while the degraded pole's threshold (1.62) is high enough that the learned arm beats it across the whole interpolation range, down to 12% of the data, and grounding reverts to helping only under scaffold *extrapolation*, where the learned constant breaks (2.6±2.0) back above the threshold—in the mean over the six splits, three of which still have grounding hurting. The clustering unit is the Hammett reaction series, of which only G=4 exist, so the bootstrap resolves nothing finer than series-level sign agreement, read as a sign test on four units (p=1/16 on the pole where grounding hurts). The construction, the estimator validation and the measured fidelity sweep are in §S7.

3.7 Scope, accuracy, and limitations

The result is a cautionary measurement about one of the two operations the word "grounding" names, and it is two measurements made in different places. On solubility: supervising a learned physical latent on reference values during training helps, and *supplying* those values at inference hurts. On infinite-dilution activity data, and there on one class of solvent: the fixed map's own error stands above the bound on what its inputs fail to carry—the condition under which better inputs surface a map's bias rather than help, and one the standard route to a better closure did not clear either. The two are measured in different regimes on largely disjoint chemistry, no measurement here connects them, and what would settle it is named in limitation (i) below and in (x); until it is run, "the substitution hurts *because* the map is misspecified" is an inference and this paper does not assert it. Both sets are scoped by the reference database that defines them and neither is the deployment regime. The useful form of the finding is one-sided and narrow: a place where grounding effort is *not* the remedy, on one row set of the map, rather than an allocation across chemistries.

The unconstrained control's representation. If the physics arm's intermediate is a surrogate, the thermodynamics might still be present implicitly in the control, which is free to represent it. It is not. A ridge probe of the control's pair representation recovers the model's own prediction at $R^2 = 0.98$ but not the measured ideal-solubility term—an upper bound at $R^2 \lesssim 0.30$ rather than an exclusion (limitation (ix)). What the representation is organised by instead is scaffold identity, consistent with the transfer limitation of §2.1.2: a test molecule on an unseen scaffold is not a small extrapolation but a point off the axis the representation is organised along. The seed-by-seed scores, the permutation nulls, the within-scaffold measure and its weakness, and the temperature probe are in §S10.

The contribution. The unit of contribution is the measurement, not any single number it returned. Its two ingredients are individually standard—the closure–insufficiency split is a law-of-total-variance identity, and the sign of grounding's effect follows the model-discrepancy principle[19,20]—and what is new is their combination into a usable instrument: an identity on a *fixed* operator, with the insufficiency side bounded rather than point-identified, whose sign rule can be tested *a priori* and *bidirectionally*. The pKa domain supplies that test as an in-distribution sign flip, at the resolution four series allow and no better—an oracle-on-a-fixed-operator affordance the concept-leakage setting lacks, its intermediate having no ground truth.[17,18] The solubility results are the instrument's first application, and their transfer out of the matched regime is not established.

Where the accuracy stands, and the confound in it. Table 5 places every arm of this work beside the external models, in three blocks that may not be read across. The physics arms trail their own control in the seed mean—the COSMO-SAC arms at every seed, the NRTL arm at two of three, being ahead at seed 42 (1.734 against 1.749; Table S2)—and the predict-the-mean row shows how little room sits above them. The physics and control arms were tuned separately and no hyperparameter-matched control was run, so the global ΔMAE of +0.14 between them—from the unrounded means, where Table 5's rounded cells differ by 0.15—carries no information about the thermodynamic bottleneck, and neither do the per-solvent-class map, the per-solute-class map, the novelty inversion or the data-efficiency trend, all of which resolve the *same* difference along chemical axes. We claim no accuracy result in either direction; those strata are reported in §S6.4 and §S6.5 as descriptions of where two particular pipelines differ, and no conclusion rests on them. Running the matched control is the direct repair and we have not run it. What survives the confound is every *within-checkpoint* contrast, where the two sides share one trained model and one tuning: §3.1, §3.2 (the ranking result included), §3.3 with the output correction folded into it, and §3.5, which uses no trained model at all. Separately, and not as an accuracy claim about the bottleneck: no configuration we ran attains a lower *seed-mean* error than the control—at one seed the NRTL arm does—and our one attempt to move the ceiling by grounding the crystal factor on external labels tested nothing, its pool being 97.76% melting points the model was already trained on (§S6.3). The one place the physical *structure* helps is temperature extrapolation, and it helps there as a property of the hypothesis class. Whether a trained arm inherits it we do not know: the only checkpoint ever scored on the temperature-withheld split is one whose predictions are near-constant, so it is untested rather than answered (§S6.1, §S5).

The external block is not like-for-like and the table says so in its own rows. FastSolv and SolProp are deposited here on a *by-solute* split, this work's arms on the harder solute-scaffold split, and the third block is as published, on other corpora and in another unit; retraining the two on the scaffold split needs a GPU run we did not perform, and scoring released FastSolv weights on our test rows is invalid because FastSolv is trained on this corpus. One sentence survives all three caveats, and it is not a favourable one: this work's control, at $\log_{10} S$ RMSE 1.00 on held-out scaffolds, is the same order as published new-solute extrapolation rather than an order away, and the physics arms are worse than every published leader at every seed and every clip. What may be read from those figures is what survives the caption's clip sweep and not the two decimals. The full reading—where each arm sits inside the sweep, which four readings fail on the seed axis as well as the clip one, the scale reference the physics arms fall under, and the doubly standardising prediction path that the re-scored rows withdraw—is in §S4.

The sign rule's own prediction. Proposition 1 predicts *when* to expect these negatives: a prior hurts precisely in the mature-data, low-noise, well-covered regime. The synthetic sweep (Fig. S9) is true by construction there, so it verifies the instrument rather than the hypothesis, while the pKa closure exercises the sufficiency channel on real data. Whether grounding helps is shaped by *two* axes, closure fidelity and how the latent is trained; characterising that plane on real systems is future work. The chemical strata form a pattern in the direction the sign rule would give, but separately tuned pipelines can produce it for reasons unrelated to the prior and there is no multiplicity control, so §S6.4 records it as the experiment a matched control should run and the diagnostic is *proposed* rather than validated.

Limitations: what the sets can carry. (i) The phenomenon and its causal attribution lie in *different regimes*: the paradox is measured where the model operates (finite composition, all γ, n=5608), the attribution and the latent-departure result only on infinite-dilution activity data (n=477 across temperatures, and the 298 K, low-γ, VT-2005-matched n=60 and n=44 sets). Transfer back to the paradox's own regime is a conjecture and is the principal open question here. The n=60 set is narrowly sourced—all by gas chromatography and all from three publications, 33 pairs from one—so neither a method-stratified nor a source-clustered re-estimate is available on it, and its margin is extraction-sensitive: a broader ThermoML pull at 298 K widens it but *inverts* it on the gas-chromatography subset, a sensitivity we leave unresolved. The broad set does not move that way under the same cut and is not the set the paradox's own oracle is defined on; the arithmetic and the enlarged-pull values are in §S9. (ii) Under the deployed convention no *aggregate* margin is resolved on either set and

Table 5: Solubility accuracy in three blocks that may not be read across: this work, external models re-run here on a different split, and published figures on other corpora. A row carries two sample sizes where its two metric columns are not scored on the same rows, one where they coincide, and a dash where the value is quoted rather than computed here.

Model	Test set	n ($\ln x_2/\log_{10} S$)	MAE $\ln x_2$	RMSE $\log_{10} S$	R^2 $\ln x_2$
This work, solute-scaffold split; mean ± s.d. over seeds, three unless the row says otherwise					
DirectGNN, same encoder, no closure *(control)*	solute scaffold, this corpus	5608 / 5440	1.70 ± 0.03	1.00 ± 0.02	0.42 ± 0.03
TGNN, NRTL closure (physics)	solute scaffold, this corpus	5608 / 5440	1.79 ± 0.07	1.54 ± 0.07	0.34 ± 0.03
TGNN, COSMO-SAC closure, supervised $\hat{\sigma}$ (physics)	solute scaffold, this corpus	5608 / 5440	1.85 ± 0.05	1.39 ± 0.19	0.33 ± 0.05
TGNN, COSMO-SAC, reference $\sigma^\star$ substituted at evaluation	solute scaffold, this corpus	5608 / 5440	2.25 ± 0.02	1.60 ± 0.14	-0.03 ± 0.04
Scale reference on the same rows					
predict the test-set mean of $\ln x_2$	solute scaffold, this corpus	5608 / 5440	2.32	1.33	0.00
External models re-run on this corpus, one run each—a different split, so not a ranking against the block above					
FastSolv, retrained on this corpus[a]	by-solute, this corpus	10287	1.44	0.84	0.55
SolProp cycle, released model + 3-parameter recalibration	by-solute, this corpus	10287	2.34	1.34	−0.18
SolProp cycle, released model, zero-shot	by-solute, this corpus	10287	2.36	1.87	−0.47
SolProp encoder retrained on $\ln x_2$, cycle discarded	by-solute, this corpus	10570 / 10287	1.62	1.07	0.39
FastSolv, retrained on this corpus[a]	random pair, this corpus	10696	1.02	0.61	0.76
SolProp encoder retrained on $\ln x_2$, cycle discarded	random pair, this corpus	10887 / 10696	0.64	0.45	0.88
As published, on their own corpora and splits; $\log_{10} S$ *in* $\mathrm{mol\,L^{-1}}$					
fastsolv (solution-FASTPROP ensemble)[1]	SolProp test set (solutes disjoint from the training corpus)	–	–	0.83	–
solution-CHEMPROP (companion)[1]	SolProp test set	–	–	0.83	–
SolProp cycle,[2] scored in[1]	SolProp test set	–	–	1.43	–
fastsolv (solution-FASTPROP ensemble)[1]	Leeds set (Boobier et al.), near room temperature	–	–	0.95	–
solution-CHEMPROP (companion)[1]	Leeds set	–	–	0.99	–
SolProp cycle,[2] scored in[1]	Leeds set	–	–	2.16	–

What the block-1 cells are. Every one is the mean over seeds of that seed's own metric, with the population standard deviation over seeds beside it; no cell pools the seeds' rows. *The unit bridge.* $\log_{10} S$ is obtained from $\ln x_2$ as $S = x_2 c_\mathrm{solv}/(1 - x_2)$, the solvent's molarity c_solv in $\mathrm{mol\,L^{-1}}$ being $1000\rho/M$ from a tabulated *mass* density ρ ($\mathrm{g\,cm^{-3}}$) and the molar mass M ($\mathrm{g\,mol^{-1}}$) where the solvent has one, and $1000/V_\mathrm{m}$ from the molar volume V_m ($\mathrm{cm^3\,mol^{-1}}$) of one generated conformer where it has none (this ρ is unrelated to the Hammett reaction constant of §3.6). Since $x_2/(1 - x_2)$ is a mole ratio, S is moles of solute per litre of *solvent* rather than of solution, and the two conventions diverge as $x_2 \to 1$. The same converter runs on every row of the first two blocks. It carries *no* temperature dependence—one ambient c_solv at every T—and it needs a molarity, so the 168 of the 5608 labelled test rows whose solvent has none carry no true $\log_{10} S$ and leave that column for *every* arm: hence the two n. Block 1's 5608 is itself the labelled subset of the 8103 test rows (Table 1); both exclusions are read off the data alone, neither off a prediction. The scale-reference row runs its one constant $\ln x_2$ prediction through the same bridge on the same 5440 rows, so it is a reference in this column too; predicting the mean of the true $\log_{10} S$ instead would read 1.28. *The clip, and what the column owes it.* Predictions are clipped at $x_2 = 0.999999$ before conversion, so an arm that saturates keeps its row with a large finite error instead of converting to $+\infty$ and being dropped. Without that clip the exclusions would differ *by arm*—12 to 195 rows per seed for the three physics arms, none for the control, which never saturates—and would read the σ-oracle arm 0.26 lower, the COSMO-SAC-grounded arm 0.31 and the NRTL arm 0.48, the control unchanged. The clip is a chosen constant and this column moves with it: swept from $x_2 = 1 - 10^{-3}$ to $1 - 10^{-10}$ it leaves the control at 1.00 throughout and carries NRTL 1.23 to 2.07, COSMO-SAC-grounded 1.18 to 1.75 and the σ-oracle 1.42 to 1.92. Two readings hold at every clip in that span, and separately at each of the three seeds: the control is below every physics arm, and every physics arm is above every published leader. Four do not, and each of the four fails on the seed axis as well as the clip one—the printed order of the NRTL and σ-oracle rows, which exchange places inside the sweep and at seed 42 but not at seeds 43 and 44; whether a physics arm beats the scale reference, which does not saturate and which they are under in 15 of the 63 arm–seed–clip cells, the deposited clip at seed 42 (1.13 against 1.33) the one such cell of the printed column; whether they sit below 1.43, the better end of the published physics-informed cycle, which at the tightest clip the NRTL and grounded arms do at every seed and the σ-oracle does in its seed mean (1.42) but not at seed 43 (1.48); and whether all three trail their control by more in this column than in MAE, which the σ-oracle does not. §S4 carries the sweep and lists every cell of the four, and §3.7 takes only the readings that survive them. [a] *Re-scored*: the same checkpoint over the same rows with the single scaling it was trained under, the doubly standardising prediction path diagnosed in §3.7 withdrawn rather than printed. §S4 gives the diagnosis, the withdrawn values and the control that reproduces them. The SolProp rows were always scored through a different path and are unaffected. The third block is quoted as published, in $\mathrm{mol\,L^{-1}}$, on other corpora and splits: no conversion of ours stands behind those six values. MAE is not available for that block, which reports RMSE and $\% \log S \pm 1$, and we do not convert between them.

only the literature-standard convention resolves one, while the headline convention is the deployed one; the aggregate falls for the stronger reason of §3.5.1 rather than for want of resolution. (iii) Of the map's six admissible cells the glycol ethers are the only row set also positive and left standing by clause (c), and not the only cell whose deletions all run; the halogenated solvents and the mono-alcohols keep their sign under the one deletion each admits, so they are unverified rather than shown to fail.

Limitations: what the estimator can deliver. (iv) B_{insuff} is upper-bounded, not point-identified (no replicates at fixed $z^\star$); the smoothness-free Jensen bound assumes m and g share the $\ln \gamma_2^\infty$ reference state, established by the closure validation of §2.3, so the mean offset is decoder bias and not a units artifact. B_{closure} is the error of a composite—the fixed map together with the calculation that produced $z^\star$—and no measurement here divides it between the two; §3.5.3 sizes that exposure and leaves the share unbounded. (v) The constant-offset margin is convention-specific and inverts under the deployed default on both sets; there the separation rests on the law-of-total-variance bound and the more-physics-worse effect. That bound is not itself convention-independent—only the estimand B_{insuff} is, Lemma 1(d)—so the deployed claim rests on it *computed under the deployed convention.* (vi) The stratification reads structure alone, so no cell was selected on its outcome, and it is exhaustive and disjoint *within* an axis but not across the two axes or their granularities: 362 of the 1711 stratum pairs share rows, all 37 water-as-solute rows lie inside three solvent classes, and 108 of the 129 alkane-solute rows are glycol-ether rows, so cells from different axes are not independent evidence and those rows are counted once. Granularity is an analyst choice, which is why three solvent levels and two solute levels are all reported. There is no multiplicity control on the per-cell bootstrap frequencies, so one cell's P describes that cell alone, and the admissibility rule is a stability filter and not a correction for the search; six of the nine solvent classes are not boundable here and three carry no estimate at all. What the search survives is measured rather than assumed, and neither test establishes the finding as a chemical effect: against 2000 chemically blind relabellings the number of admissible strata is above chance while the number of admissible *positive row sets*—the shape of the finding actually claimed—and the certified magnitude are not ($p = 0.01$, $p = 0.29$ and $p = 0.10$, none of the three signed), and the pre-declared out-of-sample test of the chemistry returned *not testable* (§3.5.1). The single finding is therefore stated at the level the instrument licenses on its own rows and *not* as a multiplicity-controlled chemical effect, and the same holds a fortiori for every cell the map reports and does not state. The rule is also one-sided in what it can deliver: with B_{insuff} bounded only from above, no cell can ever return "better inputs would help here", so the map only ever withholds a chemistry from an allocation of grounding effort.

Limitations: what is untested, and what is not auditable. (xii) The scaffold-leak guard is a property of the released stream builder and not a certified property of these runs, so the 0.20 supervision gain is the one reported number a leak could inflate and is uncertified until the leak-free re-run (§2.2); the arguments that bound the other contrasts do not restore auditability. The six remaining limitations—(vii) the accuracy ceiling against the noise floor, (viii) what the three-seed latent-departure result establishes and what it does not, (ix) the reference scarcity that bounds the black-box probe, (x) the closure-fidelity contrast and the two runs that would settle it, (xi) the separate-tuning confound stated above, and (xiii) what is checkable only by re-running the released code—are in §S6.7, with their numerals preserved.

4 Conclusions

Grounding a learned physical intermediate in reference values is two operations, not one, and only one of them helped here. Training the intermediate against reference values improved the model; supplying those values at prediction time did not. On the activity data where the two terms can be separated, they do not separate the same way in every chemistry: on one class of solvent, under a rule applied to every class alike, the bound on the fixed thermodynamic model's own error—the model as deployed, reading one computed conformer per molecule, and not its kernel alone—lies above the bound on the contribution of its inputs, which is the condition under which more accurate inputs expose a model's bias rather than remove it; on every other row set the separation is not established. *Which* class that is, this work does not establish: the class is the survivor of a search across many of them, a chemistry-blind relabelling of the same molecules returns a margin that size often enough, and the one independent database holds no rows of the chemistry the search named. The class is where the instrument answered, not where the model is shown to fail. That condition and the failed substitution are two measurements on two data sets in two regimes, so the first is offered as the reading of the second and not as a test of it. As a single number for a whole data set the comparison is an average over that set's composition, a property of the set and not of the model. And the intermediate trained through that model stopped reporting the physical quantity it is named after, on the molecules where it can be held against its reference, which is the price of reading a grounded latent as a measurement.

The general statement is a design rule with a test attached. Wherever a predictor composes a fixed physical model over a learned intermediate for which an external reference exists, that reference measures which of the two limits accuracy, and where it answers, the answer decides the remedy: a more faithful physical model—the calculation that supplies its reference values included—or better inputs. The answer comes chemistry by chemistry rather than once for a field, and it comes one-sided, so what the measurement returns is the chemistry on which better inputs are not the remedy—here one class of solvent, on which the standard route to a more faithful model did not clear the bar either, though the instrument's own null does not certify that chemistry and no out-of-sample rows of it exist. What this work establishes about the instrument is the map itself: which cells it can answer in, and at what size. Whether a latent trained through an imperfect model becomes, in general, a surrogate for that model's error is the prediction this work leaves testable.

Author contributions

N. L. Polomoshnov: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. A. V. Rudik: Writing – review & editing.

Conflicts of interest

There are no conflicts to declare.

Data availability

The analysis code, and those intermediate artifacts small enough to version, are available at `https://github.com/doctawho42/tgnn-solv` (commit **[hash to be supplied at submission]**) under the MIT licence, that commit being the one the reported numbers were produced from; the model checkpoints, the processed split, the full per-arm prediction files and the larger analysis artifacts will be deposited in a Zenodo archive, **[whose DOI is registered at submission]**. The data sources are public: BigSolDB 2.0[23] for solubility, the VT-2005 database for the reference σ-profiles, Ref.[31] as redistributed with Ref.[25] for the UD profiles, ThermoML for the infinite-dilution activity coefficients, the PGL 6th-edition IDAC database for Table 2, and the OPERA compilation[33] for the pKa data. Three deposited artifacts carry disclosed defects, recorded in §S9 and §S3: the three-seed surrogate run's per-run split provenance was not logged, one σ-stream build on disk was written against the wrong split files, and the two VT-2005-matched audit artifacts disagree on one out-of-fold estimate.

Two further defects are repaired rather than disclosed, each described where its numbers are printed: the two FastSolv retraining bundles were re-scored (§3.7, Table 5) and the withdrawn bundle is kept beside the re-scored one, and the truncated seed-44 σ-oracle per-row file has been recovered complete and now stands in the deposit (Table 3, §S9).

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Supporting Information Available

The Supporting Information is a separate document. Its sections, tables, figures and equations are numbered S1, S2, …, and every reference above that carries an S is to it. It contains supplementary methods; the remaining proofs; per-arm tables and decomposition robustness; the map's 59 strata in full, including the 44 the estimator cannot bound; further external ablations; supplementary mechanism results; supplementary diagnostics and broad negatives; the second-domain pKa construction; data provenance and reproducibility; the black-box probe tables; the synthetic closure-fidelity family; and the positioning against adjacent literatures. Two machine-readable tables are deposited with it: the 477-row broad IDAC set (Data Set S1, `broad_idac_set_477.csv`) and the 60-row VT-2005-matched set (Data Set S2, `vt2005_matched_set_60.csv`); their columns are listed in §S9.

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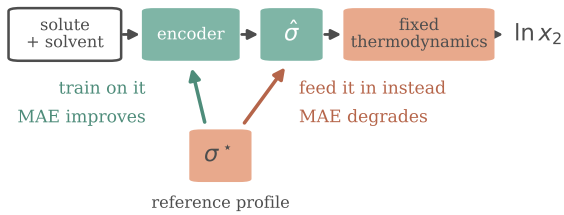
TOC Graphic



Grounding is two operations on one database, with opposite signs
(solubility MAE on held-out scaffolds, three seeds; magnitudes in the text).